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Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

A graduation research project submitted in partial fulfilment of the requirements for the degree of [Degree]

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List of Abbreviations

AI	Artificial intelligence
AOR	Adjusted odds ratio
AUC	Area under the receiver operating characteristic curve
CFA	Confirmatory factor analysis
CFI	Comparative fit index
CI	Confidence interval
CSS-12	Cyberchondria Severity Scale, 12-item short form
DWLS	Diagonally weighted least squares
ER	Emergency room
FDR	False discovery rate
GP	General practitioner
HA	Health anxiety
HC3	Heteroskedasticity-consistent standard errors, type 3
IQR	Interquartile range
KMO	Kaiser–Meyer–Olkin measure of sampling adequacy
LLM	Large language model
ML	Maximum likelihood
OLS	Ordinary least squares
OR	Odds ratio
RMSEA	Root mean square error of approximation
SD	Standard deviation
SHAI-14	Short Health Anxiety Inventory, 14-item version
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology
TLI	Tucker–Lewis index
VIF	Variance inflation factor

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Summary / Abstract

Background: Adults increasingly consult search engines and, more recently, artificial-intelligence (AI) chatbots before seeing a doctor. Health anxiety and cyberchondria (excessive, distressing online health searching) are thought to reinforce each other, but evidence from Iraq, and on AI chatbots, is scarce.

Objectives: To measure the association between health anxiety and cyberchondria among adults in Baghdad, and to describe online and AI health-information seeking and pre-consultation behaviour.

Methods: Cross-sectional online survey of 300 adults in Baghdad (July–September 2026) using the Short Health Anxiety Inventory (SHAI-14) and the Cyberchondria Severity Scale (CSS-12). Analyses included measurement-invariance testing across Arabic and English responders, FDR-corrected group comparisons, robust regression and bootstrapped mediation.

Results: Participants were mostly young (62.3% aged 20–29) and 63.3% had a health-related background; the CSS-12 was invariant across Arabic and English responders. AI chatbots were ticked by 78.0% (88.7% including respondents who named a chatbot), against 56.3% for search engines. Health anxiety correlated with cyberchondria ($r = 0.47$, 95% CI 0.38–0.55) and remained its strongest predictor after adjustment ($\beta = 0.48$), most strongly through excessive searching and post-search distress. Elevated health anxiety (SHAI-14 ≥ 18, a screening threshold rather than a diagnosis) was found in 19.0%; only 0.7% reached the stricter low-prevalence cut-off of 29. After searching, 34.7% delayed or skipped care because they felt reassured and 31.7% never told their doctor, citing fear of judgement; higher cyberchondria was associated with more disclosure. The association between AI use and cyberchondria was not robust to how AI use was defined.

Conclusion: In this convenience sample of mostly young, health-oriented adults, health anxiety and cyberchondria were linked, and AI chatbots were a major channel for online health information, without reliable evidence that their use adds to cyberchondria beyond health anxiety. The cross-sectional design cannot establish direction. Clinicians should ask about online and AI searches without judgement.

Keywords: cyberchondria; health anxiety; artificial intelligence; chatbots; online health information seeking; Iraq.

1. Introduction

1.1 Background

Searching the Internet before seeing a doctor has become an ordinary part of being ill. A symptom is now often typed into a search engine, or described to an AI chatbot, long before it is described to a physician. This shift has clear benefits: information is immediate, free and available at any hour, and it can help people judge whether a symptom needs medical attention. It also has a well-documented downside. For a sizeable minority of people, online searching does not settle a worry but amplifies it, a phenomenon that has come to be called *cyberchondria* [1,2].

Iraq offers a useful and under-studied setting in which to examine this. Internet access in the country has expanded rapidly, with roughly four in five Iraqis online by the start of 2025 [3], and in such a highly connected population the Internet and, increasingly, AI chatbots may function as a first point of contact for health concerns. How this behaviour interacts with health anxiety, and what it means for the consultation that follows, has not been examined in Baghdad.

1.2 Health anxiety

Health anxiety refers to a persistent preoccupation with having, or acquiring, a serious illness, maintained by misinterpretation of normal bodily sensations. It lies on a continuum, from transient worry that most people experience to clinically significant illness anxiety disorder and somatic symptom disorder in the DSM-5 [4]. The cognitive-behavioural model proposes that bodily sensations are interpreted as signs of serious disease and that the resulting anxiety is maintained by safety behaviours such as body checking and reassurance seeking: these behaviours relieve anxiety briefly but

prevent the person from learning that the feared outcome does not occur, so the worry returns [5].

Health anxiety is not rare. In a national Australian survey, 5.7% of adults met lifetime criteria and 3.4% current criteria for health anxiety [6], and sub-threshold forms are considerably more common. Its costs are substantial: patients in the top band of health anxiety and somatisation in primary care had markedly more physician visits and higher outpatient costs than other patients, even after adjustment for medical comorbidity [7]. In Iraq, almost half of 1591 respondents in an online survey during the COVID-19 restrictions reported health anxiety [8], and a second national online survey of 1578 Iraqi adults found that home quarantine was accompanied by greater health anxiety and increased reassurance-seeking behaviour [9], suggesting that the population is far from immune to illness worry.

1.3 Cyberchondria

The term cyberchondria was popularised by White and Horvitz, whose analysis of large-scale search logs showed that searches for common, benign symptoms frequently escalated towards rare and serious diagnoses, and that this escalation was driven by the amount and wording of medical content viewed and by the user's own predisposition to escalate [1]. Starcevic and Berle later defined cyberchondria as excessive or repeated online searching for health information that is associated with increasing, rather than relieving, distress or anxiety [2]. Crucially, searching for health information is not itself pathological; it becomes cyberchondria when it is excessive, compulsive and distressing, and when it drives further reassurance seeking [10,11]. A systematic review described cyberchondria as a possible transdiagnostic, compulsive digital syndrome closely linked to health anxiety and obsessive-compulsive symptoms, while noting that the evidence base was almost entirely cross-sectional and based on online student samples [12].

The most widely used instrument is the Cyberchondria Severity Scale (CSS), originally a 33-item measure with five factors: compulsion, distress, excessiveness, reassurance seeking and mistrust of medical professionals [13]. Its 12-item short form (CSS-12) drops the mistrust dimension and retains four three-item subscales that load substantially on a general cyberchondria factor, supporting both a total and subscale scores [14]. An Arabic version of the CSS-12 has shown acceptable structure and validity among Lebanese adults [15].

1.4 The relationship between health anxiety and cyberchondria

People with higher health anxiety search for health information more often, spend longer doing so and find searching more distressing than people with low health anxiety [16]. Among people who search online for symptoms, those with high illness anxiety tend to feel worse after searching whereas those with low illness anxiety tend to feel relieved [17]. Two meta-analyses have quantified the association: health anxiety correlates moderately with general online health information seeking ($r = 0.34$) but strongly with cyberchondria ($r = 0.62$) [18], and a later meta-analysis reported a similar pooled correlation ($r = 0.63$), with subscale correlations ranging from 0.24 to 0.66 [19]. The same work frames health-related Internet use in anxious individuals as a safety behaviour maintained by intermittent reinforcement [19]. Intolerance of uncertainty strengthens the link between searching and anxiety, presumably because search results rarely deliver a single, certain answer [20].

Whether the two conditions "fuel each other" is the central question. The only large longitudinal study, a four-wave Dutch panel of 5322 adults, found initial evidence of a reciprocal relationship in which health-anxious people search more and searching in turn raises health anxiety; however, this loop appeared among people with non-clinical health anxiety and not in the

clinically anxious subsample, in whom searching seemed to maintain rather than worsen anxiety [21]. The COVID-19 pandemic offered a natural test of the cycle: cyberchondria tracked rising concern and safety behaviours in Croatia [22], and was linked with health anxiety among university students in Turkey [23]. A recent scoping review reported cyberchondria prevalence estimates between 30.7% and 55.6% across studies and consistent associations with health anxiety [24].

1.5 From search engines to AI chatbots

Most of this evidence was gathered when "searching online" meant using a search engine. Large language model (LLM) chatbots such as ChatGPT and Gemini change the experience in important ways: they answer in fluent, personalised, conversational language; they can be questioned repeatedly; and their responses can appear more empathetic than those of physicians [25,26]. In a large US survey, about one in five respondents had used ChatGPT for health information; two-thirds of these users reported scepticism about its output and two-thirds turned to their physicians, yet about a third changed medications or requested referrals based on its answers [27]. Health anxiety has also been identified as a stable predictor of symptom-checker app use, with over half of users meeting a hypochondria classification [28]. In a representative Czech sample of 4775 Internet users, health anxiety was associated with willingness to use generative AI for health information only indirectly, through greater trust in AI [29], and among young dermatology outpatients in Turkey 61% had consulted an AI chatbot (mostly ChatGPT) before their visit, without a detectable increase in anxiety [30]. Whether AI chatbots relieve worry, by offering calm and contextualised answers, or amplify it, by being endlessly available for another round of reassurance, is an open empirical question.

1.6 Online searching and the doctor–patient relationship

What patients do after searching matters for clinical care. Online information can improve the doctor–patient relationship when patients discuss it with their physician, but can strain it when it is withheld or used to challenge the physician [31]. Reassurance seeking may translate into additional consultations, requests for investigations or "doctor shopping", while false reassurance may lead to delayed care. Studies in the Arab region have begun to describe these patterns: cyberchondria was common among adults in Jeddah and associated with smartphone addiction [32], and among Saudi primary-care patients cyberchondria was linked with health anxiety [33]. Across 13 Arabic-speaking countries of the Middle East and North Africa, about a third of 4276 migraine patients had delayed consultation or modified treatment because of online information, and higher trust in online information strongly predicted delayed consultation [34]. No comparable study has been reported from Iraq, and none, to our knowledge, has placed AI chatbots alongside search engines in this population.

1.7 Rationale

This study addresses three gaps. First, it provides, to our knowledge, the first estimate of the association between health anxiety and cyberchondria (using the CSS-12) in a Baghdad-based sample. Second, it examines AI chatbots as a distinct, and now dominant, source of online health information, rather than treating "the Internet" as one entity. Third, it links these psychological constructs to concrete pre-consultation behaviours, namely delaying care, self-medication, urgent visits, multiple consultations, disclosure of searches and trust in physicians, that matter directly to clinicians.

1.8 Aim and objectives

Aim: To study the relationship between health anxiety and cyberchondria and their correlates among adults in Baghdad who search for health information online, with particular attention to AI chatbots and pre-consultation behaviour.

Specific objectives:

1. To describe online and AI-based health-information seeking and pre-consultation behaviour among adults in Baghdad.
2. To estimate levels of health anxiety and cyberchondria and the strength of their association, overall and by cyberchondria dimension, independently of socio-demographic and digital-use factors (H1, H2); correlates of each construct were examined as secondary analyses.
3. To examine statistical pathways consistent with a mutually reinforcing cycle, and factors that might modify the association (H3).
4. To examine how health anxiety and cyberchondria relate to pre-consultation behaviour, disclosure of searches to physicians and trust in physicians (H4).

Hypotheses: (H1) Health anxiety is positively associated with cyberchondria, most strongly with its distress and excessiveness dimensions. (H2) The association is independent of socio-demographic and digital-use factors. (H3) Excessive searching statistically mediates the link between health anxiety and post-search distress. (H4) Higher cyberchondria is associated with care-seeking behaviours (urgent visits, multiple consultations) and with a more strained doctor–patient relationship. H1 and H2 were the primary, confirmatory hypotheses and H3 and H4 secondary; all other analyses, including the AI-use comparisons, item-level, moderation and sensitivity analyses, were exploratory.

2. Method

2.1 Study design and setting

This was an analytical cross-sectional study reported in line with the STROBE statement for observational studies [35] and the CHERRIES checklist for web-based surveys [36]. Data were collected online in Baghdad, Iraq, between 17 July 2026 and 11 September 2026.

2.2 Participants and sampling

A non-probability (convenience and snowball) sampling approach was used. The survey link was distributed through online channels (social media and messaging groups) whose members were based in Baghdad. Inclusion criteria were: age 18 years or older, ability to read Arabic or English, and provision of informed consent. Every completed submission was retained for analysis. No duplicate response patterns were found. [Authors to state whether Google Forms limited submissions to one per account.] Because the link spread by snowball sampling, the number of people who saw the invitation is unknown, so view and participation rates could not be computed. Three responses had been annotated by the team during data collection [authors to state the reason for each annotation]; they were kept, and their exclusion was tested in a sensitivity analysis. The resulting sample was predominantly young and from health-related fields (Section 3.1).

2.3 Sample size

The study was powered for its primary objective, the correlation between health anxiety and cyberchondria. Using Fisher's z approximation with two-sided $\alpha = 0.05$ and 80% power, a minimum of 194 participants is required to detect a small-to-moderate correlation ($r = 0.20$), and 85 to detect $r = 0.30$ [37]. Meta-analytic estimates of this association are far larger ($r \approx 0.62$)

[18,19], so these targets are conservative. For descriptive proportions, 300 participants give a margin of error of ±5.7% at 95% confidence (worst case, p = 0.5), compared with ±5% for the 385 participants of a Cochran-based calculation. The logistic model for elevated health anxiety had 57 events for 11 predictors (≈5 events per variable), which is acknowledged as a limitation of that model.

2.4 Data collection instrument

A bilingual (Arabic–English) self-administered questionnaire was built in Google Forms. All items were required, except the free-text description of chronic disease and the name of the AI chatbot used, which were conditional. The questionnaire had five parts:

1. **Consent and socio-demographic data:** confirmation of age ≥18 and consent; age group; gender; highest education; nature of study or work; diagnosed chronic medical condition (and its type).
2. **Digital health information seeking:** tools used (search engines, AI chatbots, dedicated medical platforms, social media; multiple answers); main AI chatbot; motivation for searching (multiple answers); and trust in medical information generated by online tools/AI (5-point scale).
3. **Health anxiety:** the 14-item Short Health Anxiety Inventory (SHAI-14) [38]. Each item offers four statements scored 0–3, giving a total of 0–42; higher scores indicate greater health anxiety. The SHAI has good reliability and validity in clinical and non-clinical samples [39,40], and an Arabic version has been validated in Saudi Arabia [41].
4. **Cyberchondria:** the CSS-12 [14], with items rated on a 5-point frequency scale (1 = never to 5 = always), giving a total of 12–60 and four subscales (3–15 each): Excessiveness (items 1–3), Compulsion (4–6), Distress (7–9) and Reassurance seeking (10–12).

5. **Pre-consultation behaviour and relationship with physicians:** behavioural responses after searching (multiple answers), whether search or AI results are disclosed to the physician, the perceived effect of digital information seeking on the relationship with and trust in physicians, and the language mainly relied upon while answering.

Each item was presented in English and Arabic. The research team produced the Arabic wording; its equivalence was examined empirically by comparing the psychometric behaviour of the scales between participants who relied mainly on Arabic and those who relied mainly on English (Section 2.6). For one SHAI item (item 2, noticing aches and pains) the four statements were displayed in a different order from the original scale ("as much as most people" before "less than most people"); all other SHAI and CSS items appeared in their original order. Items were scored according to the meaning of the chosen statement, following the original scoring key. The Arabic wording was produced by the research team rather than through formal cross-cultural adaptation [42], and the published Arabic versions of the CSS-12 and SHAI [15,41,43] were not used. As in the original scale, CSS item 3 refers to entering symptoms into a search engine.

2.5 Variables and operational definitions

- **Elevated health anxiety:** SHAI-14 ≥ 18, the optimal cut-off for distinguishing people with health anxiety from healthy controls in a Swedish diagnostic-accuracy study [44]. Because the same study recommends higher cut-offs in psychiatric (22) and low-prevalence (29) settings, the term "elevated" rather than "clinically relevant" is used, it is not a diagnosis, and prevalence at 15, 22 and 29 is also reported
- **High cyberchondria:** CSS-12 total in the upper tertile of the sample (> 33), since the CSS-12 has no validated clinical cut-off.

- **AI-chatbot user:** ticked "AI chatbots" as a tool or named an AI chatbot in the follow-up question. Thirty-two respondents named a chatbot without ticking the tool option; they were classified as users (the follow-up question was optional and specifically addressed to chatbot users). Because this decision affects the results, AI use is reported both ways (ticked only, and ticked or named), and all analyses involving AI use were repeated with the stricter definition.
- **Health-related background:** medical students, students in other health fields and healthcare workers.
- **Occupation:** free-text "Other" answers were recoded into the closest category (e.g. secondary-school students and school leavers as non-health students; housewives, engineers and business owners as non-health employed/other). Age groups 40–49, 50–59 and 60+ were merged (≥40) for inferential analysis because of small cell sizes.

2.6 Statistical analysis

Analyses were carried out in Python 3 (pandas, SciPy, statsmodels, pingouin, factor_analyzer and semopy). The analysis code is fully scripted and reproducible. All tests were two-sided with $\alpha = 0.05$.

- **Descriptive statistics:** frequencies and percentages for categorical variables; mean ± SD and median (IQR) for scores. Normality was assessed with the Shapiro–Wilk test, skewness and kurtosis.
- **Psychometrics:** internal consistency by Cronbach's α (with 95% CI) and McDonald's ω [45]; corrected item–rest correlations; sampling adequacy (KMO, Bartlett's test); and confirmatory factor analysis by maximum likelihood (ML), with fit judged by CFI, TLI, RMSEA and SRMR against conventional criteria (CFI/TLI ≥ 0.95, RMSEA ≤ 0.06–0.08, SRMR ≤ 0.08) [46]. Diagonally weighted least squares (DWLS) estimates on Pearson covariances are reported for comparison only, because their fit

indices are not comparable with these cut-offs. Equivalence of the Arabic and English responders was tested by comparing α values with Feldt's test [47], factor-loading similarity with Tucker's congruence coefficient [48] (an index of approximate structural similarity) and adjusted mean differences. Formal measurement invariance was then tested by multi-group CFA (ML with a mean structure), fitting configural, metric (equal loadings) and scalar (equal loadings and intercepts) models in sequence; each step was judged by the χ^2 difference test and by a change in CFI of no more than −0.01 and in RMSEA of no more than 0.015 [49,50]. Differential item functioning (DIF) was examined for every item by ordinal logistic regression of the item on the scale total, language and their interaction [51], with FDR correction across the 26 items.

- **Correlation analysis:** Pearson and Spearman correlations (bootstrapped 95% CIs) and partial correlations adjusted for covariates. Differences between dependent correlations (health anxiety with each CSS subscale) were tested with Williams' T2 test [52], as recommended by Steiger [53]. Item-level correlations were computed for every SHAI item with the CSS total and every CSS item with the SHAI total.
- **Group comparisons for every variable:** Mann–Whitney U (rank-biserial r as effect size) or Kruskal–Wallis H (ε^2 as effect size) for SHAI, CSS and each CSS subscale, with Welch's t-test/ANOVA as parametric confirmation and pairwise Mann–Whitney post-hoc comparisons with Holm adjustment. Associations of all categorical variables with elevated health anxiety and high cyberchondria were tested with Pearson χ^2 or Fisher's exact test (Monte-Carlo exact p when expected counts were small), with Cramér's V. Pairwise associations among all categorical variables were also screened.
- **Multiple testing:** because many associations were tested, p-values within each family of tests were adjusted with the Benjamini–Hochberg false

discovery rate procedure (q-values) [54]. The families were: the 162 group comparisons; the chi-square tests; the pairwise associations among categorical variables; the 40 coefficients of the behaviour models; the moderation terms; and the 168 item-level correlations. Unadjusted p-values are reported, and associations that do not survive correction are labelled as exploratory.

- **Multivariable models:** hierarchical OLS regression of CSS on socio-demographics (block 1), digital-use variables (block 2) and SHAI (block 3), with HC3 robust standard errors, standardised coefficients and VIF checks; the reverse model (SHAI on CSS); models for each CSS subscale; quantile regression (25th–90th percentiles) and a quadratic term to examine non-linearity. Logistic regression identified predictors of elevated health anxiety (AORs; apparent AUC and AUC corrected for optimism by 500 bootstrap refits) and of each pre-consultation behaviour; ordinal logistic regression was used for the ordered disclosure and relationship outcomes.
- **Mediation and moderation:** as an exploratory decomposition of the association into CSS facets, mediation models with bootstrap confidence intervals (5000 resamples) estimated how much of the link between health anxiety and post-search distress ran through excessive searching, and related pathways, including a reverse-direction model and a serial model [55]. Moderation of the SHAI–CSS association by trust, AI use, health background, gender, chronic disease, language and age was tested with interaction terms and simple slopes.
- **Sensitivity analyses:** exclusion of annotated responses; exclusion of straight-line responders (identical answers to all items of a scale); alternative SHAI cut-offs; bootstrap of the main regression coefficient; and comparison of early versus late responders as a proxy for non-response bias.

2.7 Ethical considerations

Participation was voluntary and anonymous. The first page of the form explained the purpose of the study, and electronic informed consent (with confirmation of age ≥18) was required before any question was shown; participants could withdraw at any time by closing the form. No names or contact details were collected, and data were stored securely and used only for research purposes. [Ethical approval: committee name, approval number and date to be inserted.]

3. Results

3.1 Characteristics of participants

A total of 300 adults completed the questionnaire; all gave consent and no item had missing data. The sample was young, with 187 (62.3%) aged 20–29 and 56 (18.7%) aged 18–19; 169 (56.3%) were female. Most were currently students (189 (63.0%)); medical students were the largest occupational group (132 (44.0%)), and 190 (63.3%) had a health-related background. A diagnosed chronic condition was reported by 65 (21.7%), most commonly migraine, hypertension, asthma, anaemia, psoriasis and diabetes. Most participants relied mainly on Arabic when answering (212 (70.7%)) (Table 1).

Table 1. Socio-demographic characteristics of the participants (n = 300)

Characteristic	n	%
Age group (years)		
18-19	56	18.7
20-29	187	62.3
30-39	21	7.0
40-49	25	8.3
50-59	9	3.0
60+	2	0.7
Gender		
Female	169	56.3
Male	131	43.7
Highest education		
Currently a student	189	63.0

High school or below	18	6.0
Diploma/Vocational	7	2.3
Bachelor	65	21.7
Postgraduate	21	7.0
Study/work		
Medical student	132	44.0
Other health-science student	43	14.3
Healthcare worker	15	5.0
Non-health student	46	15.3
Non-health employee/other	64	21.3
Health-related background (1=yes)		
Yes	190	63.3
No	110	36.7
Diagnosed chronic condition (1=yes)		
Yes	65	21.7
No	235	78.3
Language relied on		
Arabic	212	70.7
English	88	29.3

3.2 Patterns of online health information seeking

AI chatbots were the most frequently used tool for looking up symptoms: 234 (78.0%) ticked them, rising to 266 (88.7%) when the 32 respondents who named a chatbot without ticking the option are included. They were followed by search engines (169 (56.3%)), dedicated medical platforms (55 (18.3%)) and social media (32 (10.7%)); 171 (57.0%) used more than one type of tool (counting AI use as defined in Section 2.5). AI use was highest among the youngest participants (96% at 18–19 years) and lowest among the oldest (72% at ≥40), so the 34 non-users were older (29% aged ≥40 vs 10% of users, $p = 0.004$) but did not differ in health background (59% vs 64%). ChatGPT was the dominant chatbot (178 (59.3%)), followed by Gemini (71 (23.7%)) and Claude (12 (4.0%)). Notably, 123 (41.0%) used AI chatbots without using search engines at all. Convenience was the leading motive (80.0%), followed by deciding whether a doctor visit was needed (57.7%) and seeking instant reassurance about an unexplained symptom (51.0%). Trust in medical information generated by online tools or AI was mostly moderate (3 on the 1–

5 scale: 170 (56.7%)); only 27 (9.0%) trusted it very much or completely (Table 2, Figure 1).

Table 2. Online health information seeking, pre-consultation behaviour and relationship with physicians (n = 300)

Variable	n	%
Tools used (multiple)		
AI chatbots*	266	88.7
Search engines	169	56.3
Dedicated medical platforms	55	18.3
Social media	32	10.7
Main AI chatbot		
ChatGPT	178	59.3
Gemini	71	23.7
Claude	12	4.0
Other AI	3	1.0
None reported	36	12.0
Tool profile		
AI + search engine	143	47.7
AI without search engine	123	41.0
No AI (search/other only)	34	11.3
Motivation (multiple)		
Convenience/accessibility	240	80.0
Check if doctor visit is needed	173	57.7
Instant reassurance	153	51.0
Self-diagnose / double-check prescription	69	23.0
Cost/logistic barriers	54	18.0
Trust in online/AI medical data (1-5)		
1 – Do not trust at all	20	6.7
2 – Slightly	83	27.7
3 – Moderately	170	56.7
4 – Very much	25	8.3
5 – Completely	2	0.7
Behaviour after searching (multiple)		
Delay/skip care (reassured)	104	34.7
Ignore symptom	90	30.0
Visit multiple doctors	76	25.3
Urgent visit/ER/fast diagnosis	66	22.0
Self-medication	64	21.3
Delay care (terrified/overwhelmed)	26	8.7
Disclose search to physician		
Always	81	27.0
Sometimes	124	41.3
Never (fear of judgement)	95	31.7
Effect on doctor relationship		
No negative impact	196	65.3
Somewhat complicated	66	22.0
Significantly reduced trust	38	12.7

Tools, motivations and behaviours allowed multiple answers, so percentages do not sum to 100. AI users include 32 respondents who named a chatbot without ticking the AI option.

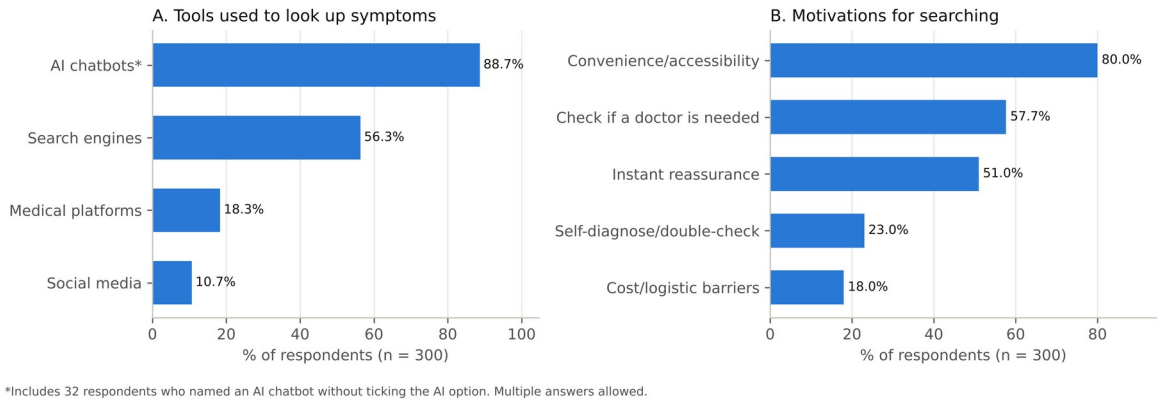


Figure 1. *Tools used and motivations for searching for health information online (n = 300).*

3.3 Pre-consultation behaviour and the doctor–patient relationship

After searching (multiple responses allowed), the most frequent response was to delay or skip professional care because the online information was reassuring (104 (34.7%)), followed by ignoring the symptom (90 (30.0%)). Behaviours suggesting increased use of services were also common: consulting multiple doctors to double-check findings (76 (25.3%)) and requesting an urgent appointment or visiting the emergency room (66 (22.0%)). Self-medication without consulting a doctor was reported by 64 (21.3%), and 26 (8.7%) delayed care because the information had frightened or overwhelmed them. Only 81 (27.0%) always disclosed their online or AI findings to their physician, whereas 95 (31.7%) never did so for fear of being judged or dismissed. Although most reported no negative effect on their relationship with physicians, 66 (22.0%) said it had become somewhat complicated and 38 (12.7%) that it had significantly reduced their trust in their doctor (Figure 2).

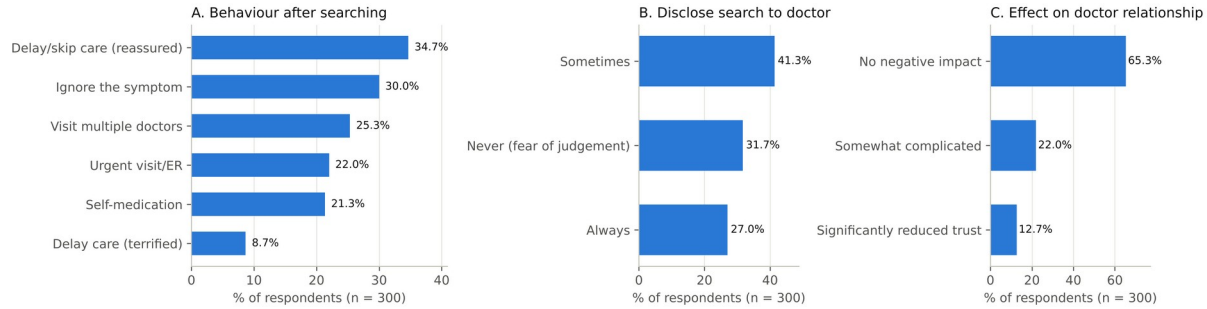


Figure 2. *Behavioural responses after searching, disclosure of searches to physicians and perceived effect on the doctor–patient relationship (n = 300).*

3.4 Levels of health anxiety and cyberchondria, and psychometric properties

The mean SHAI-14 score was 12.35 ± 5.83 (median 12, IQR 8–16). Using the cut-off of 18, 57 (19.0%) (95% CI 15.0–23.8%) had elevated health anxiety; prevalence was 32.7%, 7.3% and 0.7% at cut-offs of 15, 22 and 29, respectively, so the estimate depends heavily on the threshold chosen. The mean CSS-12 score was 28.86 ± 8.36 (median 29). Excessiveness was the highest-scoring dimension (9.66 ± 2.94) and compulsion the lowest (5.00 ± 2.58). At item level, 63.0% often or always searched online when they noticed an unexplained bodily sensation, whereas fewer than 7% reported that searching often interfered with work, leisure or social activities. Scores deviated modestly from normality (Shapiro–Wilk $p < 0.01$; skewness 0.52 for SHAI and 0.31 for CSS), so non-parametric tests were used for bivariate comparisons and robust standard errors for regression.

Both scales showed good reliability (Table 3): $\alpha = 0.81$ ($\omega = 0.82$) for SHAI-14 and $\alpha = 0.86$ ($\omega = 0.87$) for CSS-12, with subscale α between 0.76 and 0.88. Sampling adequacy was good (KMO = 0.84 and 0.85; Bartlett $p < 0.001$). Under ML, the four-factor structure of the CSS-12 was acceptable (CFI = 0.954, TLI = 0.937, RMSEA = 0.070, SRMR = 0.053) and clearly better than a single factor (CFI = 0.644, SRMR = 0.114); standardised loadings ranged from 0.61 to 0.92, and the four factors were substantially

correlated. The SHAI-14 one-factor model fitted poorly under ML (CFI = 0.830, RMSEA = 0.079, SRMR = 0.064), and a two-factor model gave little improvement, so the SHAI-14 should not be regarded as strictly unidimensional, consistent with the multi-factor solutions reported previously [39,40]. DWLS indices, which were higher (CFI ≥ 0.98), are shown in Table 3 for comparison only. Reliability was virtually identical in Arabic and English responders (SHAI α 0.81 vs 0.81, Feldt p = 0.96; CSS α 0.86 vs 0.88, p = 0.29), factor loadings were highly similar (Tucker φ = 0.99 and 0.99, indicating approximate structural similarity), and mean scores did not differ by language after adjustment (Table 3).

Formal tests of measurement invariance across the two language groups supported the equivalence of the CSS-12. Its four-factor configural model fitted acceptably in both groups (CFI = 0.937, RMSEA = 0.083); constraining loadings (metric: $\Delta\chi^2(8) = 5.52$, p = 0.70; ΔCFI = +0.002) and then intercepts (scalar: $\Delta\chi^2(8) = 8.41$, p = 0.40; ΔCFI = 0.000) did not worsen fit, indicating full scalar invariance. For the SHAI-14, whose one-factor model fitted poorly in both groups, as in the full sample (configural CFI = 0.808), loadings were invariant (metric: $\Delta\chi^2(13) = 9.61$, p = 0.72; ΔCFI = +0.004) but intercepts were not fully invariant (scalar: $\Delta\chi^2(13) = 29.0$, p = 0.007; ΔCFI = −0.018). At item level, no item showed DIF after FDR correction (smallest q = 0.13; all $\Delta R^2 \leq 0.017$); SHAI items 4, 5, 6, 8, 12 showed nominal uniform DIF only. Language also did not modify the association between health anxiety and cyberchondria (interaction B = −0.18, p = 0.29).

Table 3. Descriptive statistics, reliability and language equivalence of the SHAI-14 and CSS-12 (n = 300)

Scale	Mean ± SD	Median (IQR)	Range	α (95% CI)	ω
SHAI-14 total (0–42)	12.35 ± 5.83	12 (8–16)	1–30	0.81 (0.78–0.84)	0.82
CSS-12 total (12–60)	28.86 ± 8.36	29 (23–34)	12–57	0.86 (0.84–0.89)	0.87
Excessiveness (3–15)	9.66 ± 2.94	10 (8–12)	3–15	0.77 (0.72–0.81)	0.77

Compulsion (3–15)	5.00 ± 2.58	4 (3–6)	3–15	0.88 (0.86–0.90)	0.89
Distress (3–15)	7.24 ± 2.74	7 (5–9)	3–15	0.76 (0.71–0.81)	0.77
Reassurance (3–15)	6.96 ± 2.81	7 (5–9)	3–15	0.76 (0.71–0.81)	0.77

Language equivalence: SHAI mean 12.40±5.73 (Arabic) vs 12.23±6.08 (English), adjusted p = 0.52; CSS mean 28.96±8.21 vs 28.62±8.77, adjusted p = 0.89. CFA fit, ML (primary): CSS-12 four factors CFI 0.954, TLI 0.937, RMSEA 0.070, SRMR 0.053; SHAI-14 one factor CFI 0.830, RMSEA 0.079, SRMR 0.064. DWLS on Pearson covariances (comparison only, not comparable with ML cut-offs): CSS-12 CFI 1.000, RMSEA 0.003; SHAI-14 CFI 0.977, RMSEA 0.038. Measurement invariance across languages (multi-group CFA): CSS-12 scalar invariance supported (scalar ΔCFI = 0.000); SHAI-14 metric invariance supported, scalar not (ΔCFI = −0.018).

3.5 Association between health anxiety and cyberchondria

Health anxiety was positively and moderately correlated with cyberchondria (Pearson r = 0.47, 95% CI 0.38–0.55; Spearman ρ = 0.47, bootstrap 95% CI 0.36–0.56; $p < 0.001$), with health anxiety accounting for 22% of the variance in cyberchondria (Figure 3). The association was essentially unchanged after adjustment for age, gender, health background, chronic disease, trust in online information and AI use (partial r = 0.48, 95% CI 0.38–0.56). Health anxiety was correlated with all four cyberchondria dimensions, but not equally: correlations with excessiveness (r = 0.46) and distress (r = 0.45) were similar to each other and significantly stronger than those with compulsion (r = 0.25) and reassurance seeking (r = 0.24) (Williams' t, all $p < 0.001$) (Table 4, Figure 4).

At item level, the CSS items most strongly linked to health anxiety were panicking when reading that a symptom could signal a rare or serious condition (ρ = 0.46), searching whenever an unexplained sensation is noticed (ρ = 0.43) and feeling more anxious after searching (ρ = 0.43). Among SHAI items, fear of serious illness (ρ = 0.39) and interpretation of bodily sensations (ρ = 0.33) correlated most with cyberchondria, whereas simple awareness of bodily sensations and of aches and pains did not (ρ = 0.09 and 0.08, $q > 0.10$). The more tools and motives participants reported, the higher both scores were, whereas trust in online/AI information was associated only with excessiveness (ρ = 0.19). A full item-by-item analysis of all 168 SHAI × CSS

item pairs (Appendix, Figure 9) showed that 111 pairs remained significant after FDR correction and 58 after Holm–Bonferroni correction; the strongest were fear of serious illness with panic on reading about a rare or serious condition (ρ = 0.43), interpreting bodily sensations with searching whenever a sensation is noticed (ρ = 0.42), and difficulty distracting oneself from health thoughts with the same panic item (ρ = 0.41). Simple awareness of aches, pains and bodily sensations (SHAI items 2 and 3) was largely unrelated to any searching behaviour.

Table 4. Correlation between health anxiety (SHAI-14) and cyberchondria (CSS-12) and its dimensions (n = 300)

Cyberchondria measure	Pearson r (95% CI)	Spearman ρ	p	Adjusted β
CSS-12 total	0.47 (0.38–0.55)	0.47	<0.001	0.48
Excessiveness	0.46 (0.37–0.55)	0.45	<0.001	0.44
Compulsion	0.25 (0.14–0.36)	0.32	<0.001	0.29
Distress	0.45 (0.36–0.54)	0.44	<0.001	0.44
Reassurance	0.24 (0.13–0.35)	0.23	<0.001	0.25

Adjusted β: standardised coefficient of SHAI from robust OLS models adjusted for age, gender, occupation, chronic disease, language, tools, trust and motives. Excessiveness and Distress correlations were each significantly stronger than Compulsion and Reassurance (Williams' t, p < 0.001); they did not differ from each other (p = 0.81).

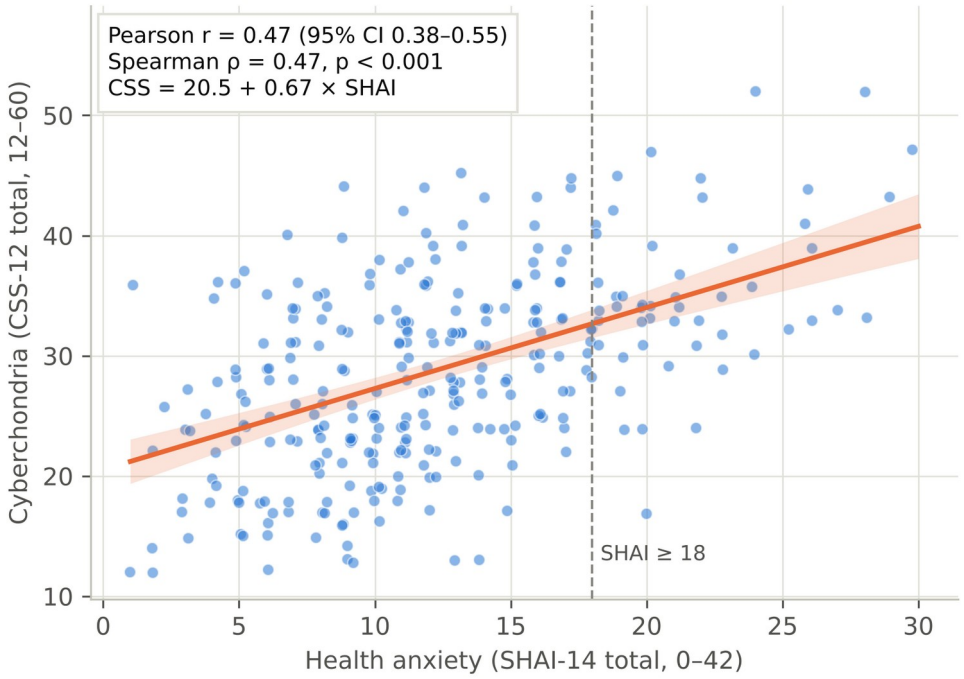


Figure 3. *Scatter plot of health anxiety (SHAI-14) against cyberchondria (CSS-12) with the least-squares line and 95% confidence band (n = 300; points slightly jittered).*

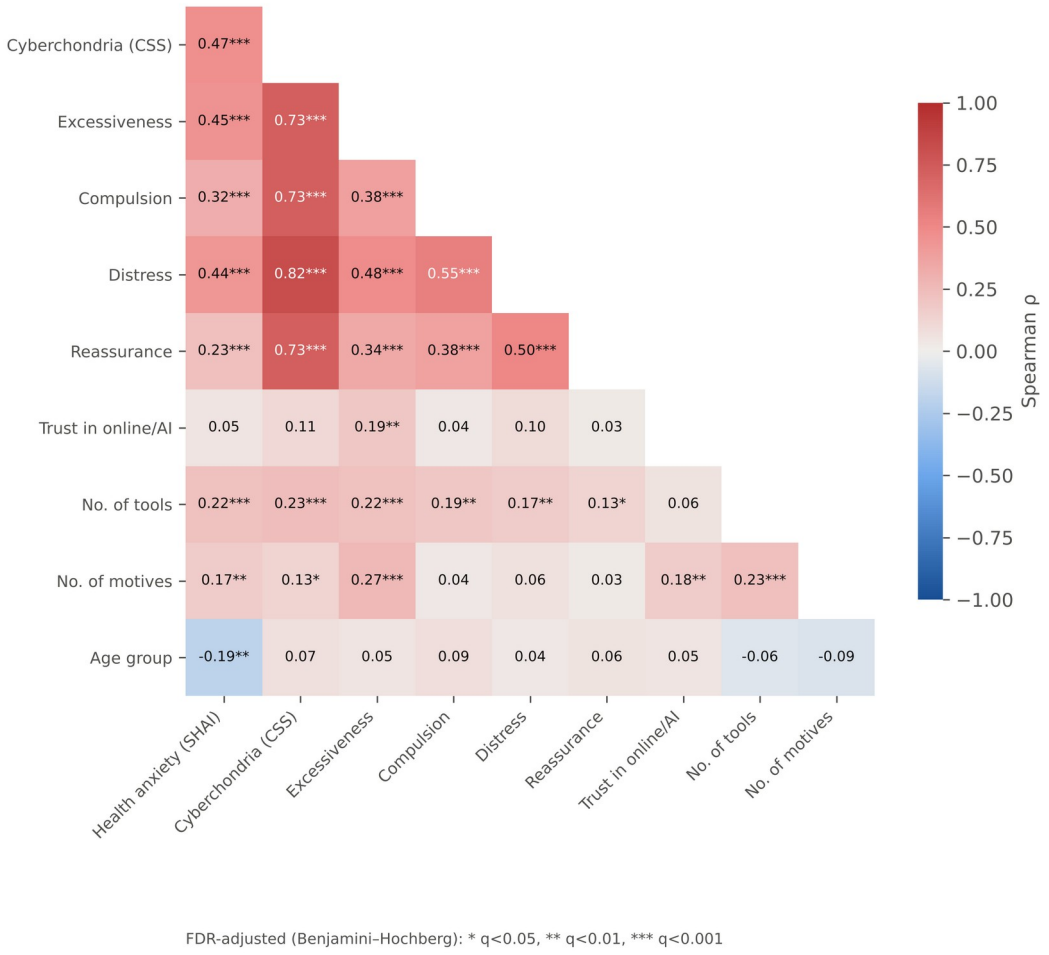


Figure 4. *Spearman correlation matrix of health anxiety, cyberchondria and its dimensions, trust in online/AI information, number of tools, number of motives and age group (FDR-adjusted significance).*

3.6 Factors associated with health anxiety and cyberchondria (bivariate analysis)

Every variable was tested against SHAI, CSS and each CSS dimension (162 tests; full results in the supplementary workbook). Table 5 summarises the main comparisons. Health anxiety tended to be higher in younger participants (18–19 years: 14.00 ± 5.49 vs 30–39 years: 10.19 ± 6.37; p = 0.008, q = 0.051), in current students (p = 0.002), in medical students compared with non-health employees (Kruskal–Wallis p = 0.005; Holm-adjusted pairwise p = 0.004), in those with a chronic condition (p = 0.004, q = 0.028), in social-media users (p = 0.001, q = 0.016) and, most clearly, in those whose motive was to obtain instant reassurance (13.76 ± 5.65 vs 10.88 ± 5.65;

$p < 0.001$, $q = 0.002$). The age difference was borderline after FDR correction ($q \approx 0.05$) and is treated as exploratory. Gender was not associated with either health anxiety ($p = 0.378$) or cyberchondria ($p = 0.859$).

Under the inclusive definition, cyberchondria was higher among AI-chatbot users than non-users (29.49 ± 8.21 vs 23.97 ± 8.05; $p < 0.001$, rank-biserial $r = 0.38$, $q = 0.005$) and differed by tool profile, rising from non-AI users to those who used AI alone and was highest in those who combined AI with search engines ($p < 0.001$; Figure 5). Of the CSS dimensions, AI use was linked with excessiveness and reassurance seeking but not with compulsion. Among chatbot users, the specific chatbot (ChatGPT, Gemini or Claude) made no difference to cyberchondria ($p = 0.726$). Participants who delayed care because online information terrified them had higher scores on every measure, particularly distress ($p < 0.001$), whereas those who decided to ignore the symptom had lower cyberchondria ($p = 0.003$). Participants who reported urgent visits had higher reassurance-seeking scores ($p = 0.002$).

Table 5. Health anxiety (SHAI-14) and cyberchondria (CSS-12) scores by participant characteristics (n = 300)

Variable	n	SHAI mean ± SD	p (q)	CSS mean ± SD	p (q)
Age group			**0.008 (0.051)**		**0.288 (0.505)**
18-19	56	14.0 ± 5.5		28.6 ± 8.1	
20-29	187	12.4 ± 5.8		28.4 ± 8.0	
30-39	21	10.2 ± 6.4		32.5 ± 11.3	
≥40	36	11.0 ± 5.7		29.7 ± 8.2	
Gender			**0.378 (0.589)**		**0.859 (0.918)**
Female	169	12.7 ± 6.0		29.0 ± 8.7	
Male	131	11.9 ± 5.5		28.6 ± 8.0	
Study/work			**0.005 (0.039)**		**0.820 (0.918)**
Medical student	132	13.3 ± 5.8		28.9 ± 8.1	
Other health-science student	43	12.2 ± 5.5		29.2 ± 9.1	
Healthcare worker	15	10.4 ± 5.1		29.5 ± 7.4	
Non-health student	46	13.0 ± 6.0		27.6 ± 7.8	
Non-health employee/other	64	10.5 ± 5.7		29.4 ± 9.1	
Chronic condition			**0.004 (0.028)**		**0.435 (0.623)**
Yes	65	14.2 ± 6.0		29.9 ± 8.9	

No	235	11.9 ± 5.7		28.6 ± 8.2	
Uses AI chatbots			**0.008 (0.051)**		**<0.001 (0.005)**
Yes	266	12.6 ± 5.7		29.5 ± 8.2	
No	34	10.5 ± 6.7		24.0 ± 8.1	
Uses search engines			**0.111 (0.316)**		**0.113 (0.316)**
Yes	169	12.8 ± 5.9		29.4 ± 7.9	
No	131	11.8 ± 5.7		28.2 ± 8.9	
Uses social media			**0.001 (0.016)**		**0.021 (0.093)**
Yes	32	15.8 ± 6.9		32.5 ± 9.1	
No	268	11.9 ± 5.6		28.4 ± 8.2	
Tool profile			**0.006 (0.039)**		**<0.001 (0.005)**
AI + search engine	143	13.1 ± 5.6		30.1 ± 7.6	
AI without search engine	123	12.0 ± 5.8		28.7 ± 8.8	
No AI (search/other only)	34	10.5 ± 6.7		24.0 ± 8.1	
Motive: reassurance			**<0.001 (0.002)**		**0.085 (0.280)**
Yes	153	13.8 ± 5.7		29.6 ± 7.6	
No	147	10.9 ± 5.7		28.1 ± 9.1	
Motive: self-diagnosis			**0.217 (0.435)**		**0.079 (0.266)**
Yes	69	13.1 ± 6.2		30.4 ± 8.1	
No	231	12.1 ± 5.7		28.4 ± 8.4	
Trust in online/AI			**0.052 (0.190)**		**0.110 (0.316)**
1	20	10.1 ± 6.4		24.8 ± 7.8	
2	83	13.2 ± 6.1		28.9 ± 8.3	
3	170	11.9 ± 5.6		28.9 ± 8.4	
4	25	14.7 ± 5.2		31.8 ± 7.1	
5	2	13.0 ± 1.4		25.5 ± 17.7	
Language			**0.857 (0.918)**		**0.654 (0.802)**
Arabic	212	12.4 ± 5.7		29.0 ± 8.2	
English	88	12.2 ± 6.1		28.6 ± 8.8	

p: Mann–Whitney U (two groups) or Kruskal–Wallis H (three or more groups); q: Benjamini–Hochberg false-discovery-rate adjusted p across all 162 group comparisons. Effect sizes, medians (IQR), Welch tests and post-hoc comparisons are provided in the supplementary workbook.

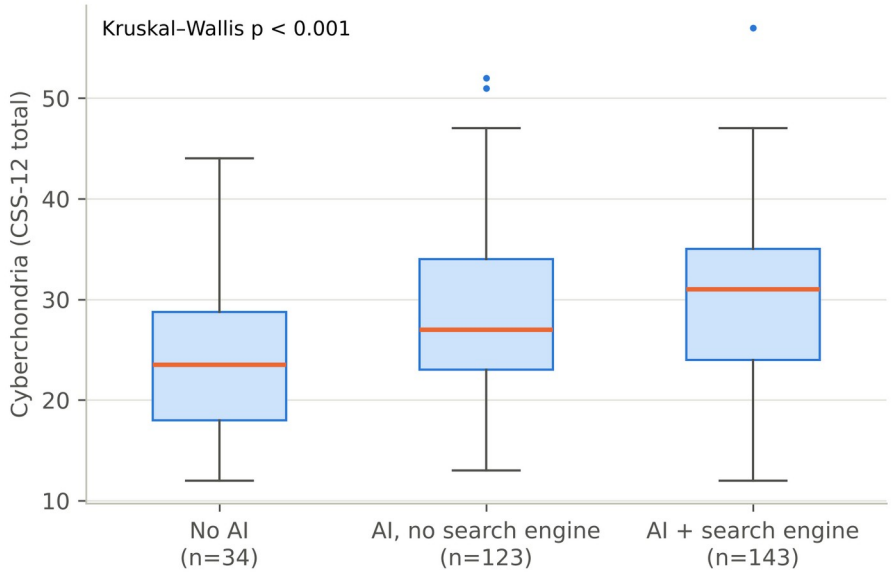


Figure 5. *Cyberchondria (CSS-12) by online tool profile.*

In categorical analysis, elevated health anxiety was more common among social-media users (13/32 (40.6%) vs 44/268 (16.4%); q = 0.015) and those seeking instant reassurance (42/153 (27.5%) vs 15/147 (10.2%); q = 0.004). Participants with elevated health anxiety had 3.7 times the odds of high cyberchondria (52.6% vs 23.0%; OR = 3.71, 95% CI 2.04–6.76). High cyberchondria was associated with disclosure pattern (p = 0.001, q = 0.015): it was most frequent among those who always disclosed their searches (33/81 (40.7%)) and least frequent among those who never did (15/95 (15.8%)) (Table 6).

Table 6. Variables significantly associated (p < 0.05) with elevated health anxiety and with high cyberchondria

Variable	n/N (%) with outcome, by group	Cramér's V	p	q (FDR)
Elevated health anxiety (SHAI ≥ 18)				
Motive: reassurance	No: 15/147 (10.2%); Yes: 42/153 (27.5%)	0.22	<0.001	0.004
Uses social media	No: 44/268 (16.4%); Yes: 13/32 (40.6%)	0.19	0.001	0.015
Delay care (terrified)	No: 47/274 (17.2%); Yes: 10/26 (38.5%)	0.15	0.016	0.107
Motive: convenience	No: 17/60 (28.3%); Yes: 40/240 (16.7%)	0.12	0.039	0.180
High cyberchondria (CSS > 33)				
Elevated health anxiety	No: 56/243 (23.0%); Yes: 30/57 (52.6%)	0.26	<0.001	0.001
Disclose to physician	Always: 33/81 (40.7%); Never : 15/95 (15.8%); Sometimes: 38/124 (30.6%)	0.21	0.001	0.015
Ignore symptom	No: 70/210 (33.3%); Yes: 16/90 (17.8%)	0.16	0.006	0.060
Uses AI chatbots	No: 3/34 (8.8%); Yes: 83/266 (31.2%)	0.16	0.007	0.060

Tool profile	AI + search engine: 49/143 (34.3%); AI without search engine: 34/123 (27.6%); No AI (search/other only): 3/34 (8.8%)	0.17	0.012	0.096
Effect on doctor relationship	No negative impact: 46/196 (23.5%); Significantly reduced trust: 14/38 (36.8%); Somewhat complicated: 26/66 (39.4%)	0.16	0.023	0.141
Main AI chatbot	ChatGPT: 61/178 (34.3%); Claude: 4/12 (33.3%); Gemini: 17/71 (23.9%); None reported: 4/36 (11.1%); Other AI: 0/3 (0.0%)	0.18	0.034	0.180
Delay care (terrified)	No: 74/274 (27.0%); Yes: 12/26 (46.2%)	0.12	0.039	0.180

Pearson χ^2 or Fisher's exact test (Monte-Carlo exact p where expected counts were < 5). Associations with $q \geq 0.05$ are exploratory. Non-significant associations are listed in the supplementary workbook.

3.7 Independent predictors (multivariable analysis)

In the hierarchical regression (Table 7), socio-demographic variables alone explained little of the variance in cyberchondria ($R^2 = 0.030$). Adding digital-use variables raised R^2 to 0.133 ($\Delta R^2 = 0.103$, $p < 0.001$), and adding health anxiety raised it to 0.317 ($\Delta R^2 = 0.184$, $p < 0.001$; adjusted $R^2 = 0.268$). Cohen's f^2 was 0.12 for the digital-use block and 0.27 for health anxiety, a medium effect. In the final model, health anxiety was by far the strongest predictor (B = 0.68, 95% CI 0.53–0.84, $\beta = 0.48$, $p < 0.001$): each additional SHAI point was associated with a 0.68-point higher CSS score. Age 30–39 years was associated with higher cyberchondria than age 20–29 ($p = 0.043$), but this was nominal and not robust. Under the inclusive definition, AI-chatbot use was associated with a 5.3-point higher CSS score (95% CI 2.34–8.32, $\beta = 0.20$, $p < 0.001$). This association was not robust: when only respondents who ticked "AI chatbots" were counted as users (n = 234), it disappeared (B = 1.49, $p = 0.314$). The 32 reclassified respondents had the highest cyberchondria of any group (mean 31.6), and the remaining non-users were a small, older group (n = 34). There is therefore no reliable evidence that AI use is associated with cyberchondria independently of health anxiety. Occupation, health background, gender, chronic disease, language, trust and motives were not independently associated with cyberchondria. Multicollinearity was low (maximum VIF = 3.9), and the bootstrap 95% CI for the SHAI coefficient (0.54–0.83) confirmed its stability. In exploratory analyses, quantile regression showed a consistent effect of health anxiety

across the distribution of cyberchondria (B from 0.55 to 0.85), and a quadratic term was not significant (p = 0.245), indicating a linear association.

In a supplementary reverse model (the same association viewed from the other side), cyberchondria was the strongest predictor of health anxiety (β = 0.45, p < 0.001; ΔR^2 = 0.173), together with chronic disease (B = 1.68, p = 0.029) and searching for instant reassurance (B = 2.42, p < 0.001); after adjustment, medical students no longer differed from other groups (p = 0.101).

Table 7. Hierarchical robust linear regression of cyberchondria (CSS-12 total) – final model (n = 300)

Predictor	**B**	**95% CI**	**β**	**p**
Constant	12.04	6.39–17.68		<0.001
Age 18–19 (ref. 20–29)	−0.43	−2.77–1.91	−0.02	0.721
Age 30–39	**6.27**	**0.19–12.34**	**0.19**	**0.043**
Age ≥40	3.98	−0.86–8.81	0.16	0.107
Healthcare worker (ref. non-health employed)	1.06	−4.32–6.44	0.03	0.700
Medical student	0.41	−3.67–4.49	0.02	0.845
Non-health student	−0.73	−5.12–3.66	−0.03	0.745
Other health-science student	1.32	−3.61–6.25	0.06	0.599
Female	−0.05	−1.82–1.72	0.00	0.954
Chronic condition	−0.14	−2.42–2.15	−0.01	0.906
Answered mainly in English	0.51	−1.71–2.74	0.03	0.650
Uses AI chatbots	**5.33**	**2.34–8.32**	**0.20**	**<0.001**
Uses search engines	0.95	−1.00–2.90	0.06	0.342
Uses medical platforms	0.17	−2.10–2.44	0.01	0.885
Uses social media	1.46	−1.61–4.54	0.05	0.351
Trust in online/AI (per point)	0.72	−0.50–1.93	0.06	0.247
Motive: instant reassurance	−1.11	−2.91–0.69	−0.07	0.229
Motive: self-diagnosis	0.99	−1.01–3.00	0.05	0.332
Motive: cost/logistics	0.44	−1.50–2.39	0.02	0.656
Motive: need for a doctor?	0.09	−1.71–1.90	0.01	0.919
Health anxiety (SHAI, per point)	**0.68**	**0.53–0.84**	**0.48**	**<0.001**

Block 1 (socio-demographics) R^2 = 0.030; Block 2 (+ digital use) R^2 = 0.133, ΔR^2 = 0.103, p = 0.0002; Block 3 (+ SHAI) R^2 = 0.317, ΔR^2 = 0.184, p < 0.001; adjusted R^2 = 0.268; F = 8.03, p < 0.001. HC3 robust standard errors. β: standardised coefficient.

In logistic regression (Table 8, Figure 6), elevated health anxiety was independently associated with higher cyberchondria (AOR = 3.12 per SD, 95% CI 2.09–4.65), with searching for instant reassurance (AOR = 3.83, 95% CI 1.75–8.42) and with younger age (AOR = 0.61 per age-group step, 95% CI 0.38–0.96). The apparent AUC was 0.83; after bootstrap correction for optimism (500 resamples) it was 0.79 (McFadden pseudo-R^2 = 0.22). With

about five events per predictor, the estimates, particularly the wide-interval odds ratio for the reassurance motive, are likely to be optimistic and are treated as exploratory.

Table 8. Multivariable logistic regression of elevated health anxiety (SHAI ≥ 18; 57 events, n = 300)

Predictor	AOR	95% CI	p
Age group (per step)	**0.61**	**0.38–0.96**	**0.034**
Female	1.23	0.63–2.40	0.551
Health-related background	1.05	0.47–2.35	0.909
Chronic condition	1.51	0.64–3.53	0.345
Answered mainly in English	0.94	0.45–1.97	0.871
Uses AI chatbots	0.61	0.22–1.69	0.346
Uses search engines	1.43	0.69–2.96	0.330
Trust in online/AI (per point)	0.74	0.46–1.21	0.232
Motive: instant reassurance	**3.83**	**1.75–8.42**	**<0.001**
Motive: self-diagnosis	0.93	0.42–2.09	0.867
Cyberchondria (per 1 SD)	**3.12**	**2.09–4.65**	**<0.001**

AUC = 0.83 (apparent; optimism-corrected 0.79); McFadden pseudo-R^2 = 0.22; likelihood-ratio p < 0.001; robust (HC1) standard errors.

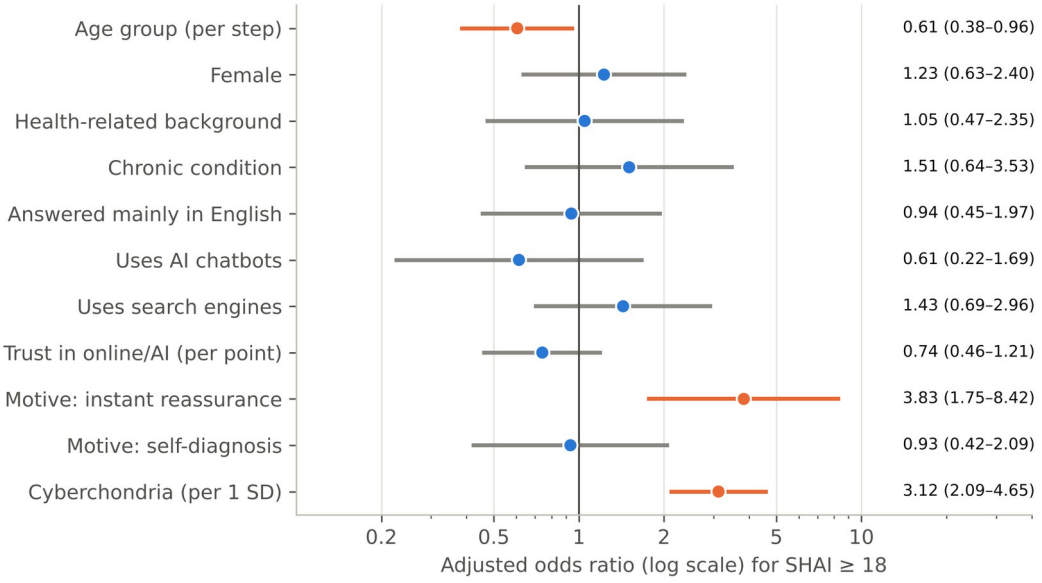


Figure 6. *Adjusted odds ratios for elevated health anxiety (SHAI ≥ 18). Orange: p < 0.05.*

3.8 Health anxiety, cyberchondria and pre-consultation behaviour

When health anxiety and cyberchondria were entered together, each with adjustment for trust, AI use and socio-demographics (Table 9), cyberchondria, rather than health anxiety, was associated with behaviours: each SD increase in CSS raised the odds of an urgent visit or ER attendance (AOR = 1.51, 95%

CI 1.09–2.10) and of delaying care because of fear (AOR = 1.59, 95% CI 1.09–2.30), and lowered the odds of ignoring the symptom (AOR = 0.65, 95% CI 0.48–0.89). Higher cyberchondria was also associated with a more strained relationship with physicians in the ordinal model (OR = 1.34, 95% CI 1.02–1.77); in the binary model of any negative effect (Table 9) the estimate was similar (AOR = 1.35, 95% CI 1.00–1.83, p = 0.049), a nominal result that is not robust. For disclosure, higher cyberchondria decreased the odds of never telling the physician about online searches (AOR = 0.57, 95% CI 0.43–0.78, q = 0.012), the only behavioural association to survive FDR correction. Higher health anxiety showed the opposite tendency (AOR = 1.47, 95% CI 1.08–1.98, p = 0.013), but this did not survive correction (q = 0.098) and is exploratory. Because the response option combined frequency with a reason, "never" here means never disclosing, citing fear of judgement. Greater trust in online/AI information was associated with lower odds of consulting multiple doctors (AOR = 0.59, 95% CI 0.41–0.84) and showed a non-significant trend towards delaying care because of reassurance (AOR = 1.33, p = 0.088). Neither score was associated with self-medication or with delaying care because of reassurance. Apart from the cyberchondria–disclosure association, these behavioural associations did not survive FDR correction across the 40 tests (q between 0.08 and 0.28) and should be regarded as exploratory.

Table 9. Adjusted associations of health anxiety and cyberchondria with pre-consultation behaviours and the doctor–patient relationship

Outcome	Events	SHAI AOR (95% CI)	p	CSS AOR (95% CI)	p
Urgent visit/ER	66	0.91 (0.66–1.25)	0.551	1.51 (1.09–2.10)	0.014
Multiple doctors	76	1.06 (0.79–1.43)	0.688	1.20 (0.90–1.60)	0.208
Delay/skip care (reassured)	104	0.89 (0.66–1.19)	0.417	0.83 (0.63–1.10)	0.190
Delay care (terrified)	26	1.49 (0.94–2.36)	0.087	1.59 (1.09–2.30)	0.015
Self-medication	64	0.81 (0.56–1.16)	0.245	1.00 (0.71–1.41)	0.994
Ignore symptom	90	1.16 (0.85–1.58)	0.359	0.65 (0.48–0.89)	0.007
Never disclose to physician	95	1.47 (1.08–1.98)	0.013	0.57 (0.43–0.78)	<0.001
Negative effect on doctor relationship	104	1.03 (0.78–1.38)	0.824	1.35 (1.00–1.83)	0.049

AOR per 1 SD of each score, from logistic models containing both scores plus trust, AI use, age, gender, health background, chronic disease and language. FDR-adjusted q-values are given in the supplementary workbook.

3.9 Pathways: mediation and moderation

As an exploratory decomposition of the association into CSS facets (standardised, adjusted, 5000 bootstrap resamples; Table 10, Figure 7), excessive searching statistically accounted for part of the association between health anxiety and post-search distress (indirect effect = 0.16, 95% CI 0.11–0.23; 35% of the total effect), with the direct effect remaining significant (c′ = 0.30). Post-search distress statistically accounted for almost all (98%) of the association between health anxiety and reassurance-seeking consultations (indirect = 0.22, 95% CI 0.16–0.29; direct c′ = 0.01, p = 0.932). In a serial model, the indirect path from health anxiety through excessive searching and distress to reassurance seeking was also significant (indirect = 0.07, 95% CI 0.04–0.11). The reverse ordering, from excessive searching through distress to health anxiety, was also significant (indirect = 0.15, 95% CI 0.09–0.22), so the data cannot distinguish direction; because mediators and outcomes are subscales of one instrument, these estimates also partly reflect overlapping item content. Participants with elevated health anxiety scored higher on all four dimensions, most markedly on excessiveness and distress (Figure 8).

None of the tested moderators significantly altered the strength of the SHAI–CSS association after FDR correction (Table 10): trust in online/AI information (interaction p = 0.34), AI use (p = 0.47), health background (p = 0.66), gender (p = 0.98), chronic disease (p = 0.75) and language (p = 0.29); there was a non-significant trend towards a weaker association with increasing age (p = 0.068). Consistently, the correlation was moderate to strong in every subgroup with adequate size, ranging from r = 0.35 to 0.57; it was somewhat stronger in medical students (r = 0.57) than in other participants (r = 0.40; z-test p = 0.056).

Table 10. Exploratory mediation models decomposing the association between health anxiety (HA) and the dimensions of cyberchondria (standardised coefficients, n = 300)

Model	a	b	Total (c)	Direct (c ′)	Indirect (95% boot CI)
A. HA → excessive searching → post-search distress	0.46	0.35	0.46	0.30	0.16 (0.11–0.23)
B. HA → excessive searching → compulsion	0.46	0.27	0.28	0.15	0.13 (0.07–0.20)
C. HA → post-search distress → reassurance seeking	0.46	0.48	0.22	0.01 (ns)	0.22 (0.16–0.29)
D. Excessive searching → distress → HA (reverse)	0.49	0.30	0.47	0.32	0.15 (0.09–0.22)
E. Serial: HA → excess → distress → reassurance				−0.04 (ns)	0.07 (0.04–0.11)

a: predictor → mediator; b: mediator → outcome adjusted for the predictor; indirect effect = a × b = c − c′. All models adjusted for age, gender, health background, chronic disease, trust and AI use; 5000 bootstrap resamples. All a, b and total paths p < 0.001. Moderation of the SHAI–CSS association (interaction terms): trust in online/ai data p = 0.34; ai chatbot use p = 0.47; health-related background p = 0.66; female gender p = 0.98; chronic condition p = 0.75; english-language responder p = 0.29; age group (ordinal) p = 0.07 (all q > 0.40).

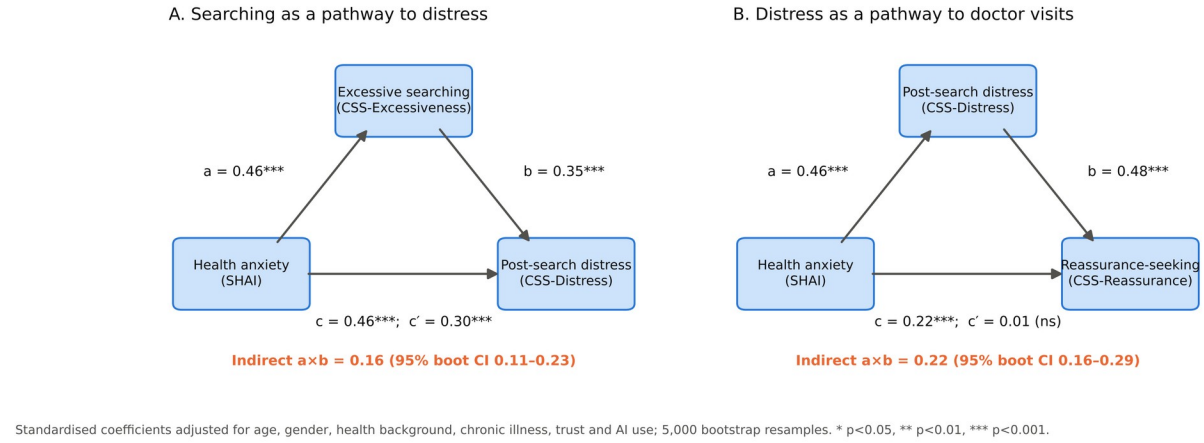


Figure 7. *Mediation models: (A) excessive searching as a pathway from health anxiety to post-search distress; (B) post-search distress as a pathway from health anxiety to reassurance seeking.*

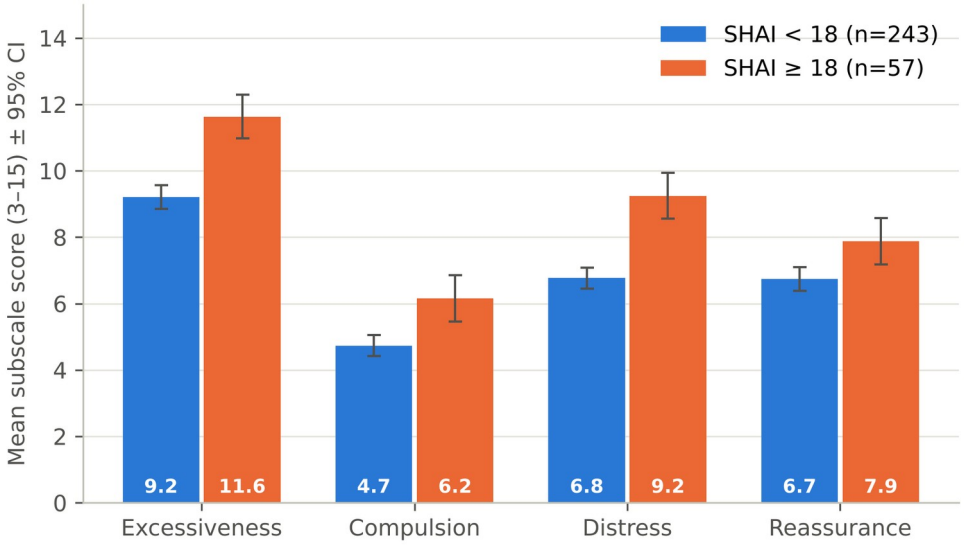


Figure 8. *Cyberchondria dimensions by health-anxiety status (mean ± 95% CI).*

3.10 Sensitivity analyses

Results were robust. Excluding the three annotated responses (r = 0.46) or the 9 straight-line responders (identical answers to every item of the SHAI-14 or of the CSS-12; n = 291, r = 0.47) left the main correlation unchanged. Dropping the one SHAI item whose response options were reordered (item 2) also left it unchanged: using the 13-item total (range 0–39), r = 0.48, identical when prorated to the 0–42 range. Early and late responders, a weak proxy for non-response bias, did not differ in cyberchondria (p = 0.46) or health anxiety (p = 0.07). The regression residuals showed mild non-normality and heteroskedasticity (Shapiro–Wilk p = 0.020; Breusch–Pagan p = 0.004), which is why robust standard errors and bootstrap intervals were used; conclusions did not change.

4. Discussion

4.1 Principal findings

This study is, to our knowledge, the first to apply the CSS-12 in a Baghdad-based sample, and among the first to place AI chatbots alongside search engines as a source of health information before consulting a doctor.

Four findings stand out. First, AI chatbots were used by 78.0–88.7% of participants, depending on the definition, more than search engines, and about two in five used AI without search engines at all. Second, health anxiety and cyberchondria were moderately and robustly associated (r = 0.47), independently of all other measured factors, and health anxiety was the dominant predictor of cyberchondria. Third, the association was concentrated in the excessiveness and distress dimensions; an exploratory decomposition showed that most of it ran statistically through excessive searching and post-search distress. Fourth, cyberchondria was reflected in behaviour towards physicians, most robustly in more frequent disclosure of searches.

4.2 Health anxiety and cyberchondria

The observed correlation (r = 0.47) supports H1 and H2 but is weaker than the pooled estimates of about 0.62–0.63 from meta-analyses [18,19]. Part of the gap reflects measurement error: corrected for the reliability of both scales, the correlation would be about 0.56, close to the meta-analytic range, although our observed estimate remains 0.47. Other explanations are also plausible. Our sample had relatively low health anxiety (mean SHAI 12.3) and a narrow range in which correlations are attenuated; recruitment through online channels may have favoured confident, habitual Internet users for whom searching is routine rather than alarming; and most previous studies used the full 33-item CSS, whose distress and reassurance items overlap more with health anxiety. The pattern across dimensions is also informative. Distress and excessiveness, the dimensions closest to the cognitive model of health anxiety, were most strongly related to SHAI, while compulsion (interference with work and social life) was rare and only weakly related, consistent with the subscale range reported previously [19]. At item level, it was the *interpretation* of sensations and the *fear* of serious illness, not mere awareness of bodily sensations, that tracked cyberchondria. This is consistent

with the view that cyberchondria is more closely tied to catastrophic appraisal than to bodily vigilance itself [5,20].

The prevalence of elevated health anxiety (19% at SHAI ≥ 18) should be interpreted cautiously. The cut-off was derived against healthy controls in Sweden [44]; it is not a diagnosis, and prevalence ranged from 7% to 33% across plausible cut-offs. Nevertheless, it indicates that a substantial minority of digitally active adults in Baghdad experience health worry at a level that would warrant clinical attention, consistent with high rates of health anxiety reported in Iraq during the pandemic [8].

4.3 Do health anxiety and cyberchondria fuel each other?

The mediation results are consistent with the reinforcing cycle proposed by cognitive-behavioural models and by the longitudinal Dutch study [5,21]: people with higher health anxiety search more, more searching goes with more distress, and distress goes with more reassurance seeking from doctors. That the direct effect of health anxiety on reassurance seeking became negligible once post-search distress was accounted for is compatible with online searching being one route by which health anxiety is turned into healthcare use, but it may equally reflect the strong overlap between the Distress and Reassurance subscales. However, these results must not be over-interpreted. In cross-sectional data, mediation tests are tests of a hypothesised ordering, not of causation, they can be seriously biased estimates of longitudinal processes [56], and a significant indirect effect cannot by itself identify the true mediator or the causal order [57]. Our reverse-direction model (searching → distress → health anxiety) was equally significant, which is exactly what one would expect if the relationship is bidirectional, but also what one would expect if both constructs simply share variance. In addition, the mediators and outcomes in models A–C are subscales of the same instrument, so part of their association reflects shared method variance and

overlapping item content [58]. The appropriate conclusion is that our data are *compatible with*, but cannot *demonstrate*, a cycle in which health anxiety and cyberchondria fuel each other.

H3 is therefore supported statistically, but not causally.

4.4 The role of AI chatbots

The extent of AI chatbot use (78% ticked; 88.7% including those who named a chatbot) is a new finding and contrasts with a 2024 US survey in which about one in five respondents reported using ChatGPT for health information [27], and with a 2025 online survey in which 21% used LLM chatbots for health information against 98% using search engines [59]. Our sample was young and highly educated, which partly explains the difference: AI use fell from 96% in those aged 18–19 to 72% in those aged 40 and over. Under the inclusive definition, AI users had higher cyberchondria, mainly in excessiveness and reassurance seeking, but this association disappeared under the stricter definition, rested on a small and older comparison group (n = 34), and AI use did not modify the strength of the health anxiety–cyberchondria link. We therefore find no reliable evidence that AI chatbot use is associated with cyberchondria independently of health anxiety; this is an absence of evidence rather than evidence of absence. Because LLM chatbots answer in fluent, personalised and empathetic language [25,26], they could invite repeated, conversational querying; alternatively, people who already search heavily may simply adopt every available tool. The level of AI use in our sample resembles that of young Turkish dermatology outpatients (61%), in whom pre-visit AI consultation mainly confirmed an existing decision to seek care and was not associated with increased anxiety [30]; in a large Czech sample, health anxiety was linked to AI use only through trust [29], whereas in our data trust was unrelated to health anxiety. On current evidence, AI chatbots appear to be a new channel for, rather than a demonstrated driver of,

cyberchondria, which is consistent with observations that health-anxious people are over-represented among users of digital symptom checkers [28].

Trust in online and AI information was mostly moderate and was associated with excessive searching but not with distress. Higher trust was also associated with lower odds of consulting several doctors, suggesting that trusted AI answers may substitute for some "doctor shopping". This is potentially beneficial for health services, but it is also where false reassurance becomes a risk: higher trust showed a non-significant trend towards delaying care because of reassurance (AOR = 1.33, p = 0.088), echoing a multinational study from the Middle East and North Africa in which trust in online information strongly predicted delayed consultation [34].

4.5 Pre-consultation behaviour and the doctor–patient relationship

The most common response after searching, delaying or skipping care because the information was reassuring (35%), is a double-edged finding. For self-limiting symptoms it may save unnecessary visits; for serious ones it represents a risk of delayed diagnosis. Notably, this behaviour was not related to health anxiety or cyberchondria, meaning it occurs across the spectrum and cannot be dismissed as a feature of anxious users. Conversely, cyberchondria was associated with urgent visits and emergency attendance and with fear-driven delay, although these associations did not survive FDR correction and are exploratory, mirroring evidence that health anxiety increases healthcare utilisation [7]. Thus online searching appears to push different people in opposite directions: towards complacency in some and towards over-use of services in others.

The disclosure findings are clinically important. Almost a third of participants said they never told their doctor about their searches for fear of being judged or dismissed; because the response option combined frequency with a reason, this figure describes non-disclosure attributed to fear of

judgement rather than all non-disclosure. Higher cyberchondria was robustly associated with more disclosure ($q = 0.012$), whereas the tendency for higher health anxiety to go with concealment was nominal only ($q = 0.098$). One interpretation is that patients for whom searching is a dominant behaviour bring their findings to the consultation as part of reassurance seeking, while health-anxious patients may anticipate being labelled as "worried well"; the second part of this interpretation needs confirmation. Concealment matters because online information discussed openly can strengthen the doctor–patient relationship, whereas undisclosed information is more likely to undermine it [31]. One in three participants reported that searching had complicated (22%) or significantly reduced (13%) their relationship with or trust in physicians, and this tended to be more likely at higher levels of cyberchondria (a nominal association).

H4 is therefore only partially supported: the only behavioural association that survived correction for multiple testing was between higher cyberchondria and more frequent disclosure of searches to physicians.

4.6 Medical and health-science students

Medical students had higher unadjusted health anxiety than non-health employees, but this difference disappeared after adjustment, largely reflecting their younger age; age was an independent predictor of elevated health anxiety. The SHAI–CSS correlation tended to be stronger among medical students. This is compatible with the idea of "medical student syndrome", but evidence for that concept is weak [60], and our data suggest that medical knowledge does not protect against cyberchondria (medical students' CSS scores were identical to those of other participants).

4.7 Comparison with regional studies

Our findings are in line with emerging Arab-region data. In Jeddah, cyberchondria was common and associated with smartphone addiction [32]; among Saudi primary-care patients, cyberchondria had a positive effect on health anxiety in structural models [33]; and the Arabic CSS-12 showed a structure similar to ours in Lebanon [15]. Among 1435 Egyptian medical students, the population closest to ours, 85.8% showed moderate cyberchondria on the CSS-12, and worry about one's own or a family member's health predicted it [61]; a Saudi validation of the CSS-12 likewise found acceptable reliability and positive correlations with health anxiety [43]. The good fit of the four-factor CSS-12, its full scalar invariance across Arabic and English responders and the absence of item-level DIF add to the evidence that the instrument functions well in Arabic-speaking populations.

4.8 Strengths and limitations

Strengths of this study include the use of two established instruments with good reliability and confirmed structure; a bilingual questionnaire with formal tests of measurement invariance across languages; the separate assessment of AI chatbots; complete data; correction for multiple testing; robust regression, bootstrap inference and several sensitivity analyses; and the linking of psychological measures to concrete behaviours relevant to clinical practice.

The limitations are substantial and should be weighed carefully:

- **Design and causality:** the cross-sectional design cannot establish temporal order; mediation results describe statistical pathways, not causal mechanisms, and may be biased relative to longitudinal effects.
- **Sampling and generalisability:** an online convenience/snowball sample is subject to self-selection. The sample was young, highly educated and

over-represented medical students (44.0%) and people with health backgrounds (63.3%), so prevalence estimates and the frequency of AI use cannot be generalised to the adult population of Baghdad. Residence in Baghdad was assured by the distribution channels rather than verified by a question, and non-Internet users, older adults and less-educated people are under-represented by design.

- **Measurement:** the Arabic wording was produced by the research team without a formal forward–backward translation and cognitive pre-testing; formal tests supported full (scalar) invariance of the CSS-12 and metric invariance of the SHAI-14 across language groups, with no item showing DIF after correction, but SHAI-14 intercepts were not fully equivalent, the language groups were self-selected rather than randomised, and statistical invariance does not replace formal cross-cultural adaptation with cognitive pre-testing. Published Arabic versions of both scales exist [15,41,43] but were not used, so comparability with Arabic studies that used them is uncertain. On the form, the response options of one SHAI item (item 2) were displayed out of their original order; scoring followed the original key, and dropping the item did not change the main result, but the order may still have influenced answers. CSS item 3 refers only to search engines and may under-capture repeated querying of AI chatbots. "Language relied on" was self-selected, not randomised. The SHAI-14 one-factor model fitted poorly under maximum-likelihood estimation. The SHAI cut-off was derived in Sweden, and the CSS-12 has no validated clinical cut-off, so the "high cyberchondria" group is sample-relative.
- **Single-item and self-designed measures:** trust, motivations, behaviours, disclosure and relationship were assessed with single, non-validated items, some allowing multiple answers. The questionnaire asked for "most frequent" behaviours but allowed several answers; the trust item combined online tools and AI, so it cannot separate trust in search results from trust

in chatbots; the disclosure option combined frequency with a reason (fear of judgement); and "somewhat complicated" is not clearly a negative effect on the relationship.

- **Self-report bias:** all measures were self-reported and may be affected by recall and social desirability bias; no clinical data or diagnoses were available, and chronic disease was self-reported.
- **Multiple testing and power:** despite FDR correction, the large number of exploratory tests means that some isolated associations may be false positives; several behavioural associations were nominally significant only. Some subgroups (non-AI users, n = 34; healthcare workers, n = 15; older age groups) were small, and the logistic model for clinical health anxiety had about five events per predictor.
- **Classification of AI use:** 32 respondents named an AI chatbot without ticking the AI option. They were classed as users, and the independent association between AI use and cyberchondria depended on this decision, so it should be regarded as tentative.
- **Data quality:** an online survey cannot exclude careless responding entirely; nine straight-line responders were identified, although their exclusion did not change the results.

5. Conclusions and Recommendations

5.1 Conclusions

1. AI chatbots, mainly ChatGPT, have become a leading source of health information before consulting a doctor among digitally active adults in Baghdad: 78.0% ticked them (88.7% including those who named a chatbot), more than used search engines.
2. About one in five participants (19%) had elevated health anxiety (SHAI ≥ 18, a screening threshold rather than a diagnosis; 0.7% at the stricter

cut-off of 29), and younger people and those searching for instant reassurance were most affected.

3. Health anxiety and cyberchondria are moderately and robustly associated ($r = 0.47$); health anxiety is the strongest independent predictor of cyberchondria, especially of excessive searching and post-search distress.
4. The pattern of associations is compatible with a reinforcing cycle (health anxiety, excessive searching, distress, reassurance-seeking consultations), but the cross-sectional decomposition, based on subscales of one instrument, cannot establish direction.
5. There is no reliable evidence that AI-chatbot use is associated with cyberchondria independently of health anxiety: the association depended on how inconsistent answers about AI use were classified, and AI use did not change the strength of the health anxiety–cyberchondria link.
6. Online searching shapes the consultation: about one-third of participants delay care because they feel reassured, about one-third never disclose their searches to their doctor for fear of judgement, and about one-third say searching has complicated or reduced their trust in physicians. Higher cyberchondria was robustly linked with more disclosure; links with urgent care-seeking, concealment by health-anxious participants and a strained relationship were exploratory.

5.2 Recommendations

For clinicians and medical education:

- Routinely and non-judgementally ask patients what they have searched for or asked an AI chatbot, to bring hidden information into the consultation and reduce concealment.
- Recognise repeated searching, post-search distress and repeated reassurance seeking as possible markers of health anxiety, and screen with

a brief tool such as the SHAI where appropriate; offer cognitive-behavioural strategies rather than repeated reassurance or investigations.

- Include digital health communication, AI literacy and the management of health anxiety in undergraduate medical curricula, and support medical students who may themselves be vulnerable.

For the Ministry of Health and public-health bodies:

- May consider campaigns on digital and AI health literacy that explain both the uses and the limits of online information, especially false reassurance that leads to delayed care.
- May consider providing or endorsing reliable Arabic-language health information sources and triage services so that people searching for "whether I need a doctor" are directed to safe advice.

For developers of AI tools:

- May consider designing health responses that recognise signs of health anxiety (e.g. repeated questions about the same symptom), avoid escalating rare diagnoses without context, and encourage appropriate professional consultation, in Arabic as well as English.

For future research:

- Develop and test interventions: ClinicalTrials.gov lists only nine registered studies on cyberchondria (October 2026), none in Iraq and six of them observational and three interventional [62], so trials of brief digital health-literacy or cognitive-behavioural interventions in Arabic are a clear gap.
- Conduct longitudinal or experience-sampling studies to test whether health anxiety and cyberchondria reinforce each other over time, and experimental studies comparing AI chatbot and search-engine answers on post-search anxiety.
- Use probability sampling of the general Baghdad population, including older and less-educated adults, and use formally adapted Arabic

instruments (the published CSS-12-Ar and Arabic SHAI, or a new forward–backward adaptation with measurement-invariance testing), with CSS item 3 worded to cover both search engines and AI chatbots.

- Link self-reported behaviours to objective healthcare utilisation and clinical outcomes, especially delayed diagnosis following false reassurance.

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Appendices

Appendix A. Item-by-item correlations between the SHAI-14 and CSS-12

Item-by-item Spearman correlations between all 14 SHAI-14 items and all 12 CSS-12 items (168 pairs). Correction for multiple testing was applied within this family of 168 tests: 111 pairs were significant after Benjamini–Hochberg FDR correction and 58 after Holm–Bonferroni correction. Full values are given in the supplementary workbook (sheet T28_item_by_item).

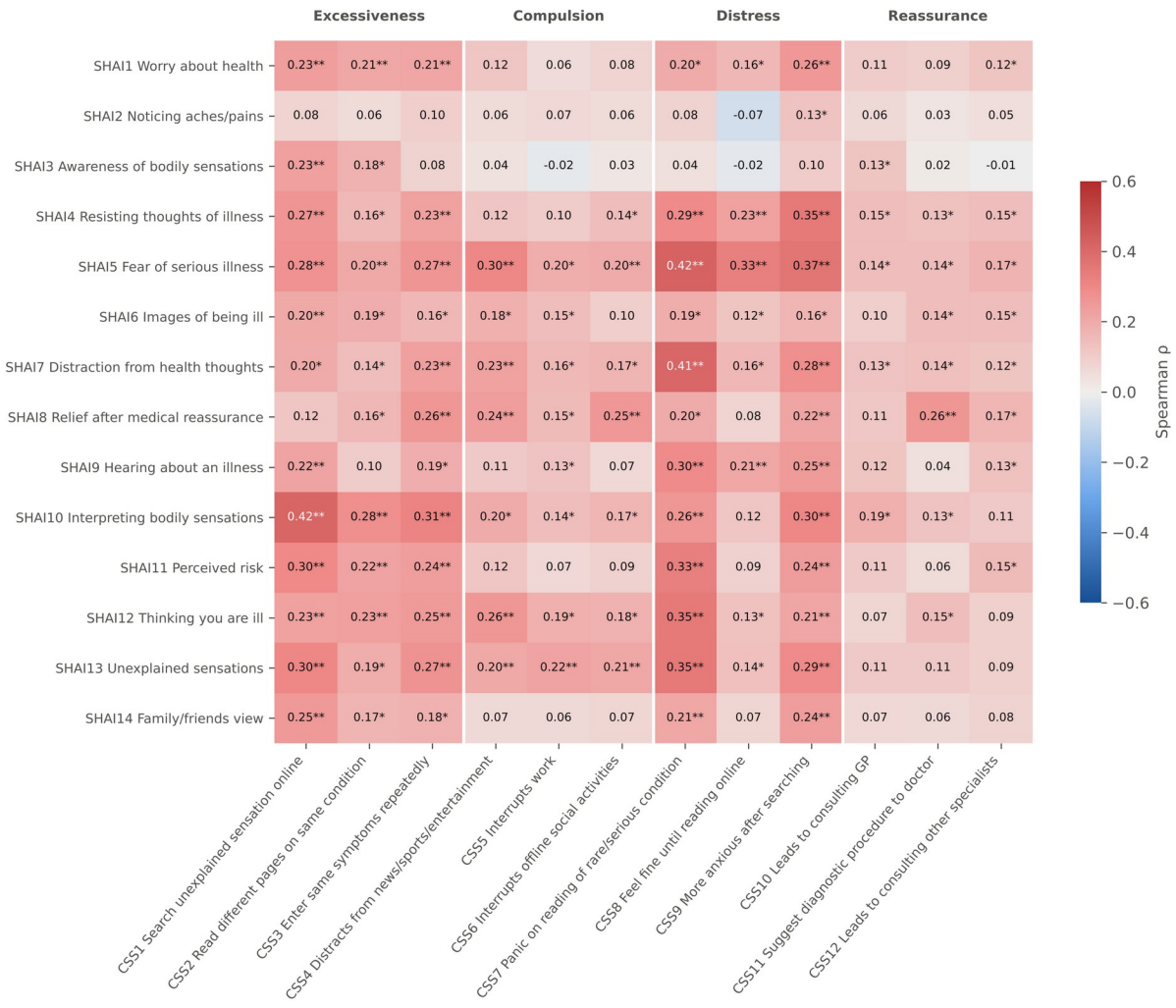


Figure 9. *Item-by-item correlations between SHAI-14 and CSS-12 items (Spearman ρ). * significant after FDR correction across the 168 pairs; ** also significant after Holm–Bonferroni correction.*

Appendix B. Study questionnaire

The bilingual (English–Arabic) questionnaire as administered through Google Forms is reproduced below (41 questions).

Digital Health Information-Seeking, Health Anxiety, and Cyberchondria Questionnaire
استبيان البحث عن المعلومات الصحية الرقمية، والقلق الصحي، والسايبركوندريا (توهم المرض عبر الإنترنت)

* Indicates required question



1. **By checking the box below, I confirm that I am at least 18 years of age, have read the description of this study, and voluntarily agree to participate. I understand that I can withdraw from this survey at any point without penalty.** *

بالتأشير على المربع أدناه، أؤكد أن عمري لا يقل عن 18 عاماً، وأنني قرأت وصف هذه الدراسة، وأوافق طواعية على المشاركة فيها. كما أفهم أنه يمكنني الانسحاب من هذا الاستبيان في أي وقت دون أي تبعات.

Mark only one oval.

I explicitly agree and consent to participate in this study. / أوافق وأقبل المشاركة في هذه الدراسة البحثية.

2. **Age Group / الفئة العمرية** *

Mark only one oval.

- 18-19
- 20-29
- 30-39
- 40-49
- 50-59
- 60+

3. **Gender: الجنس** *

Mark only one oval.

- Male / ذكر
- Female / انثى

4. **What is the highest level of education you have completed?** *

ما هو أعلى مستوى تعليمي أكملته؟

Mark only one oval.

- Currently a student / لا أزال طالباً مستمراً بالدراسة
- Postgraduate degree (Master's or Ph.D.) / دراسات عليا (ماجستير أو دكتوراه)
- Bachelor's degree / بكالوريوس
- Diploma / Vocational training // دبلوم (معهد / تدريب مهني)
- High school or below / إعدادية أو أقل
- Other: _____

5. **What is the nature of your current study or work?** *

ما هي طبيعة دراستكِ أو عملكِ الحالي؟

Mark only one oval.

- Medical student (Medicine) / طالب في كلية الطب البشري
- Student in another healthcare field (Pharmacy, Dentistry, Nursing, Analysis, Medical Technologies, etc.) / طالب في مجال صحي آخر (صيدلة، أسنان، تمريض، تحليلات، تقنيات، وما شابه)
- Medical Doctor / Physician (General Practitioner, Resident, or Specialist) / طبيب بشري (ممارس، مقيم، أو اختصاص)
- Other healthcare professional (Pharmacist, Dentist, Nurse, Technician, etc.) / عامل في القطاع الصحي غير الأطباء (صيدلاني، طبيب أسنان، ممرض، تقني، وما شابه)
- Student in a non-healthcare field / طالب في تخصص غير صحي
- Employed/working in a non-healthcare field / موظف أو عامل في مجال غير صحي
- Other: _____

6. **Do you suffer from any diagnosed chronic medical condition** *
هل تعاني من اي امراض طبية مزمنة تم تشخيصها مسبقا

Mark only one oval.

- Yes / نعم
- No / لا

7. **If you answerd yes above please specify which chronic disease you have (Leave blank if you prefer not to say)**
اذا اجبت بنعم اعلاه يرجى كتابة اي مرض مزمن لديك (اتركه فارغا اذا كنت تفضل عدم الذكر)

_____

8. **Which tools do you use most frequently when looking up health or medical symptoms online? (Select all that apply)** *
ما هي الادوات التي تستخدمها بشكل متكرر عند البحث عن الأعراض الصحية أو الطبية عبر الإنترنت؟ (يمكن اختيار اكثر من اجابة)

Tick all that apply.

- [ ] Search engine / محركات البحث (google)
- [ ] AI Chatbots / روبوتات الذكاء الاصطناعي
- [ ] Dedicated Medical Platforms / المنصات الطبية المتخصصة(WebMed, Mayo Clinic..)
- [ ] Social Media Networks / منصات التواصل الاجتماعي (Facebook ,Reddit ,Instagram)

9. **If you checked AI above**
specify which AI do you use most of the time
اذا قمت بتحديد الذكاء الاصطناعي
قم بتحديد اي برنامج غالبا

Mark only one oval.

- ChatGPT
- Gemini
- Claude
- Other:

10. **What is your primary motivation for searching medical information online? (Select all that apply):** *
ما هو دافعك الرئيسي للبحث عن معلومات طبية عبر الإنترنت؟
(يمكنك اختيار أكثر من إجابة)

Tick all that apply.

- [ ] Immediate convenience and easy accessibility / السهولة الفورية وإمكانية الوصول السريع في أي وقت
- [ ] High costs or logistical difficulty of professional doctor visits / التكاليف المرتفعة أو الصعوبات اللوجستية المرتبطة بزيارة الطبيب المختص
- [ ] Seeking instant reassurance regarding an unexplained bodily symptom / الرغبة في الحصول على طمأنينة فورية بشأن عرض جسدي غير مفسر
- [ ] Intending to self-diagnose or double-check a physician's prescription / نية التشخيص الذاتي أو التحقق من دقة وصفة الطبيب المعالج
- [ ] I just want to see if the symptom is serious enough to warrant a doctor's visit in the first place. / أريد فقط معرفة ما إذا كانت الأعراض خطيرة بدرجة كافية تستدعي زيارة الطبيب في المقام الأول
- [ ] Other: _____

11. **On a scale from 1 to 5, how much do you trust the medical data generated by online tools/AI?** *

على مقياس من 1 إلى 5، ما مدى ثقتك في البيانات الطبية الناتجة عن أدوات الإنترنت أو الذكاء الاصطناعي؟

Mark only one oval.

- Do not trust at all / لا اثق على الاطلاق
- Slightly trust/ واثق قليلا
- Moderately trust/ واثق بدرجة معتدلة
- Very much trust/ واثق جداً
- Completely trust / ثقة مطلقة

12. **Worrying about health** *
القلق بشأن الصحة

Mark only one oval.

- I do not worry about my health /أنا لا أقلق بشأن صحتي
- I occasionally worry about my health / ينتابني القلق أحياناً بشأن صحتي
- I spend much of my time worrying about my health / أقضي الكثير من وقتي في القلق بشأن صحتي
- I spend most of my time worrying about my health / أقضي معظم وقتي في القلق بشأن صحتي

13. **Body awareness / الوعي بآلام الجسد** *

Mark only one oval.

- I notice aches/pains as much as most other people (of my age) / ألاحظ الأوجاع/الآلام بنفس القدر مثل معظم الأشخاص الآخرين (من هم في عمري)
- I notice aches/pains less than most other people (of my age) / ألاحظ الأوجاع/الآلام بشكل أقل من معظم الأشخاص الآخرين (من هم في عمري)
- I notice aches/pains more than most other people (of my age) / ألاحظ الأوجاع/الآلام بشكل أكثر من معظم الأشخاص الآخرين (من هم في عمري)
- I am aware of aches/pains in my body all the time / أنا واعي ومدرك لوجود أوجاع/آلام في جسدي طوال الوقت

14. **Reaction to bodily sensations / رد الفعل تجاه الأحاسيس الجسدية** *

Mark only one oval.

- As a rule, I am not aware of bodily sensations or changes / كقاعدة عامة، أنا لا أنتبه للأحاسيس أو التغيرات الجسدية
- Sometimes I am aware of bodily sensations or changes / في بعض الأحيان أكون واعياً للأحاسيس أو التغيرات الجسدية
- I am often aware of bodily sensations or changes / غالباً ما أكون واعياً للأحاسيس أو التغيرات الجسدية
- I am constantly aware of bodily sensations or changes / أنا واعي ومدرك للأحاسيس أو التغيرات الجسدية باستمرار وبشكل دائم

15. **Thoughts of illness / الأفكار المتعلقة بالمرض** *

Mark only one oval.

- Resisting thoughts of illness is never a problem / لا تشكل مقاومة الأفكار المتعلقة بالمرض أي مشكلة بالنسبة لي على الإطلاق
- Most of the time I can resist thoughts of illness / في معظم الأوقات، يمكنني مقاومة الأفكار المتعلقة بالمرض
- I try to resist thoughts of illness but am often unable to do so / أحاول مقاومة الأفكار المتعلقة بالمرض ولكنني غالباً ما أكون عاجزاً عن ذلك
- Thoughts of illness are so strong that I no longer even try to resist them / الأفكار المتعلقة بالمرض قوية جداً لدرجة أنني لم أعد أحاول مقاومتها

16. **Fear of serious illness / الخوف من الإصابة بمرض خطير** *

Mark only one oval.

- As a rule, I am not afraid that I have a serious illness / كقاعدة عامة، لا يخالجني الخوف من أنني مصاب بمرض خطير
- I am sometimes afraid that I have a serious illness / ينتابني الخوف أحياناً من أنني مصاب بمرض خطير
- I am often afraid that I have a serious illness. / غالباً ما ينتابني الخوف من أنني مصاب بمرض خطير
- I am always afraid that I have a serious illness / أنا دائم الخوف من أنني مصاب بمرض خطير

17. **Images of illness / الصور الذهنية عن المرض** *

Mark only one oval.

- I do not have images (mental pictures) of myself being ill / لا تتكون لدي صور (تخيلات ذهنية) لنفسي وأنا مريض
- I occasionally have images of myself being ill / تتكون لدي أحياناً صور لنفسي وأنا مريض
- I frequently have images of myself being ill / تتكون لدي كثيراً صور لنفسي وأنا مريض
- I constantly have images of myself being ill / تتكون لدي باستمرار صور لنفسي وأنا مريض

18. **Distraction from health thoughts / تشتيت الانتباه عن الأفكار الصحية** *

Mark only one oval.

- I do not have any difficulty taking my mind off thoughts about my health / لا أجد أي صعوبة في صرف ذهني عن الأفكار المتعلقة بصحتي
- I sometimes have difficulty taking my mind off thoughts about my health / أجد أحياناً صعوبة في صرف ذهني عن الأفكار المتعلقة بصحتي
- I often have difficulty taking my mind off thoughts about my health / غالباً ما أجد صعوبة في صرف ذهني عن الأفكار المتعلقة بصحتي
- Nothing can take my mind off thoughts about my health / لا يوجد شيء على الإطلاق يمكنه صرف ذهني عن الأفكار المتعلقة بصحتي

19. **Medical reassurance / الطمأنينة الطبية** *

Mark only one oval.

- I am lastingly relieved if my doctor tells me there is nothing wrong / أشعر بارتياح دائم إذا أخبرني الطبيب أنه لا يوجد خطب ما
- I am initially relieved, but the worries sometimes return later / أشعر بالارتياح في البداية، لكن المخاوف تعود أحياناً لاحقاً
- I am initially relieved, but the worries always return later / أشعر بالارتياح في البداية، لكن المخاوف تعود دائماً لاحقاً
- I am not relieved if my doctor tells me there is nothing wrong / لا أشعر بالارتياح حتى لو اخبرني الطبيب انه لا يوجد خطب ما

20. **Hearing about illness / السماع عن الأمراض** *

Mark only one oval.

- If I hear about an illness, I never think I have it myself / إذا سمعت عن مرض ما، لا يخطر ببالي أبداً أنني مصاب به
- If I hear about an illness, I sometimes think I have it myself / إذا سمعت عن مرض ما، يخطر ببالي أحياناً أنني مصاب به
- If I hear about an illness, I often think I have it myself / إذا سمعت عن مرض ما، غالباً ما يخطر ببالي أنني مصاب به
- If I hear about an illness, I always think I have it myself / إذا سمعت عن مرض ما، يخطر ببالي دائماً أنني مصاب به

21. **Interpretation of sensations / تفسير الأحاسيس الجسدية** *

Mark only one oval.

- If I have a bodily sensation or change, I rarely wonder what it means / إذا شعرت بإحساس أو تغير جسدي، فنادراً ما أتساؤل عما يعنيه
- If I have a bodily sensation or change, I often wonder what it means / إذا شعرت بإحساس أو تغير جسدي، فغالباً ما أتساؤل عما يعنيه
- If I have a bodily sensation or change, I always wonder what it means / إذا شعرت بإحساس أو تغير جسدي، فأنا دائماً أتساؤل عما يعنيه
- If I have a bodily sensation or change, I must know what it means / إذا شعرت بإحساس أو تغير جسدي، فلا بد لي من معرفة ماذا يعني حتماً

22. **Perceived risk / تقدير خطر الإصابة** *

Mark only one oval.

- I usually feel at very low risk for developing a serious illness / أشعر عادةً بأن احتمالية إصابتي بمرض خطير منخفضة جداً
- I usually feel at fairly low risk for developing a serious illness / أشعر عادةً بأن احتمالية إصابتي بمرض خطير منخفضة إلى حد ما
- I usually feel at moderate risk for developing a serious illness / أشعر عادةً بأن احتمالية إصابتي بمرض خطير متوسطة
- I usually feel at high risk for developing a serious illness / أشعر عادةً بأن احتمالية إصابتي بمرض خطير عالية جداً

23. **Thinking you are ill / الاعتقاد بالمرض** *

Mark only one oval.

- I never think I have a serious illness / لا أظن أبداً أنني مصاب بمرض خطير
- I sometimes think I have a serious illness / أظن أحياناً أنني مصاب بمرض خطير
- I often think I have a serious illness / غالباً ما أظن أنني مصاب بمرض خطير
- I usually think that I am seriously ill / أظن عادةً أنني مريض جداً بالفعل

24. **Unexplained sensations / أحاسيس غير مفسرة** *

Mark only one oval.

- If I notice an unexplained bodily sensation, I don't find it difficult to think about other things / إذا لاحظت إحساساً جسدياً غير مفسر، لا أجد صعوبة في التفكير بأشياء أخرى
- If I notice an unexplained bodily sensation, I sometimes find it difficult to think about other things / إذا لاحظت إحساساً جسدياً غير مفسر، أجد أحياناً صعوبة في التفكير بأشياء أخرى
- If I notice an unexplained bodily sensation, I often find it difficult to think about other things / إذا لاحظت إحساساً جسدياً غير مفسر، أجد غالباً صعوبة في التفكير بأشياء أخرى
- If I notice an unexplained bodily sensation, I always find it difficult to think about other things / إذا لاحظت إحساساً جسدياً غير مفسر، أجد دائماً صعوبة في التفكير بأشياء أخرى

25. **Friends & family perspective / وجهة نظر العائلة والأصدقاء** *

Mark only one oval.

- My family/friends would say I do not worry enough about my health / قد يقول عائلتي/أصدقائي إنني لا أقلق كفاية بشأن صحتي
- My family/friends would say I have a normal attitude to my health / قد يقول عائلتي/أصدقائي إن لدي موقفاً طبيعياً ومتوازناً تجاه صحتي
- My family/friends would say I worry too much about my health / قد يقول عائلتي/أصدقائي إنني أقلق أكثر من اللازم بشأن صحتي
- My family/friends would say I am a hypochondriac / قد يقول عائلتي/أصدقائي إنني مصاب بـ "توهم المرض" (وسواس مرضي)

26. **If I notice an unexplained bodily sensation, I search for it on the internet.** *

إذا لاحظتُ عرضاً جسدياً غير مفسر، أقوم بالبحث عنه عبر الإنترنت.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

27. **I read different web pages about the same health condition.** *

أقرأ صفحات ويب مختلفة ومتعددة حول نفس الحالة الصحية التي أشعر بها

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

28. **I enter the same symptoms into a search engine on more than one occasion.** *

أدخل نفس الأعراض في محرك البحث في أكثر من مناسبة أو بشكل متكرر.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

29. **Researching health-related conditions or symptoms online distracts me from reading news, sports, or entertainment materials.** *

البحث عن الأعراض أو الحالات الطبية عبر الإنترنت يشتتني عن متابعة الأخبار أو الرياضة أو المواد الترفيهية.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

30. **Researching health-related conditions or symptoms online interrupts my work (e.g. working on emails or documents).** *

البحث عن الأعراض أو الحالات الطبية عبر الإنترنت يقاطع عملي (مثل كتابة رسائل البريد الإلكتروني أو العمل على المستندات).

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

31. **Researching health-related conditions or symptoms online interrupts my offline social activities (e.g. decreasing time spent with friends/family).** *

البحث عن الأعراض أو الحالات الطبية عبر الإنترنت يقاطع أنشطتي الاجتماعية الواقعية (مثل تقليل الوقت المستقطع مع الأصدقاء أو العائلة).

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

32. **I panic when I read online that a symptom I have could be a sign of a rare or serious condition.** *

أشعر بالذعر عندما أقرأ عبر الإنترنت أن العرض الذي أعاني منه قد يكون مؤشراً لحالة نادرة أو خطيرة.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

33. **I feel fine until I read about a serious medical condition online.** *

أشعر بأنني بخير إلى أن أقرأ عن حالة صحية خطيرة عبر الإنترنت.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

34. **I feel more anxious or distressed after researching potential health-related conditions or symptoms online.** *

أشعر بمزيد من القلق أو الضيق بعد البحث عن الأعراض أو الحالات الطبية المحتملة عبر الإنترنت.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

35. **Researching health-related conditions or symptoms online leads me to consult with my general practitioner.** *

البحث عن الأعراض أو الحالات الطبية عبر الإنترنت يدفعني لاستشارة طبيبي العام.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

36. **I suggest to my doctor/medical professional that I may need a specific diagnostic procedure that I have read about online (e.g. a biopsy, a specific blood test).** *

أقترح على طبيبي/المختص الصحي أنني قد أحتاج إلى إجراء تشخيصي محدد قرأت عنه عبر الإنترنت (مثل خزعة أو تحليل دم معين).

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

37. **Researching health-related conditions or symptoms online leads me to consult with other medical specialists.** *

البحث عن الأعراض أو الحالات الطبية عبر الإنترنت يدفعني لاستشارة أخصائيين طبيين آخرين.

Mark only one oval.

- Never / أبداً
- Rarely / نادراً
- Sometimes / أحياناً
- Often / غالباً
- Always / دائماً

38. **After searching your health symptoms online, what are your most frequent behavioral responses? (Select all that apply):** *

بعد البحث عن أعراضك الصحية عبر الإنترنت، ما هي استجاباتك السلوكية الأكثر تكراراً؟ (يمكنك اختيار أكثر من إجابة):

Tick all that apply.

- [ ] I purchase self-prescribed medications or treatments without consulting a doctor / أقوم بشراء أدوية أو علاجات ذاتية دون استشارة الطبيب بناءً على نتائج البحث
- [ ] I delay or completely skip seeking professional medical help because the online data reassured me / أقوم بتأجيل أو إلغاء طلب المساعدة الطبية المتخصصة تماماً لأن المعلومات المتوفرة على الإنترنت طمأنتني.
- [ ] I delay professional help because the online data terrified or overwhelmed me / أقوم بتأجيل طلب المساعدة الطبية المتخصصة لأن المعلومات المتوفرة على الإنترنت أرعبتني أو أشعرتني بالقلق الشديد
- [ ] I immediately request an urgent appointment, go to the emergency room, or seek a fast diagnosis / أطلب فوراً موعداً عاجلاً، أو أذهب إلى ردهة الطوارئ، أو أبحث عن تشخيص سريع.
- [ ] I arrange visits with multiple different doctors to double-check my findings / أقوم بزيارة عدة أطباء مختلفين للتأكد من صحة استنتاجاتي (ما عثرت عليه من معلومات)
- [ ] Decided to ignore the symptom entirely. / أقرر تجاهل الغرض تماماً

39. **During a consultation, do you explicitly share the online search results or AI outputs with your physician?** *

أثناء الاستشارة الطبية، هل تخبر طبيبك صراحةً بنتائج بحثك على الإنترنت أو بما قدمه لك الذكاء الاصطناعي؟

Mark only one oval.

- Yes, I always share my online findings transparently / نعم، أطلع الطبيب دائماً على نتائج بحثي عبر الإنترنت بكل شفافية.
- Sometimes, depending on how the doctor responds or listens / أحياناً، اعتماداً على كيفية استجابة الطبيب أو مدى إصغائه لي.
- No, I never mention my online searches out of fear of being judged or dismissed / لا، لا أذكر أبداً ما بحثت عنه عبر الإنترنت خوفاً من أن يُحكم علي (بالوسوسة مثلاً) أو أن يتم تجاهل كلامي.

40. **How has digital health information-seeking affected your relationship and trust with professional healthcare providers?** *

كيف أثر بحثك عن المعلومات الصحية الرقمية على علاقتك بمقدمي الرعاية الصحية المتخصصين (الاطباء) ومدى ثقتك بهم

Mark only one oval.

- It significantly reduced my trust in my doctor's diagnosis or treatment recommendations / قلل بشكل كبير من ثقتي في تشخيص طبيبي أو التوصيات العلاجية والدوائية المقدمة لي
- It somewhat complicated my relationship, causing me to question choices, but I still follow advice / عقد علاقتي بالطبيب إلى حد ما، مما جعلني أشكك في خياراته، لكني ما زلت ملتزماً باتباع نصيحته في النهاية
- It had no negative impact; it serves purely as helpful supplementary information to my doctor's care / لم يكن له أي تأثير سلبي؛ فهو يعمل كمجرد معلومات تكميلية مفيدة وداعمة للرعاية التي يقدمها طبيبي

41. **Which language did you rely on most while completing this questionnaire?** * **اللغة التي اعتمدت عليها بشكل أساسي أثناء تعبئة هذا الاستبيان؟**

Mark only one oval.

- Mostly English / بشكل أساسي اللغة الإنجليزية
- Mostly Arabic / بشكل أساسي اللغة العربية

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